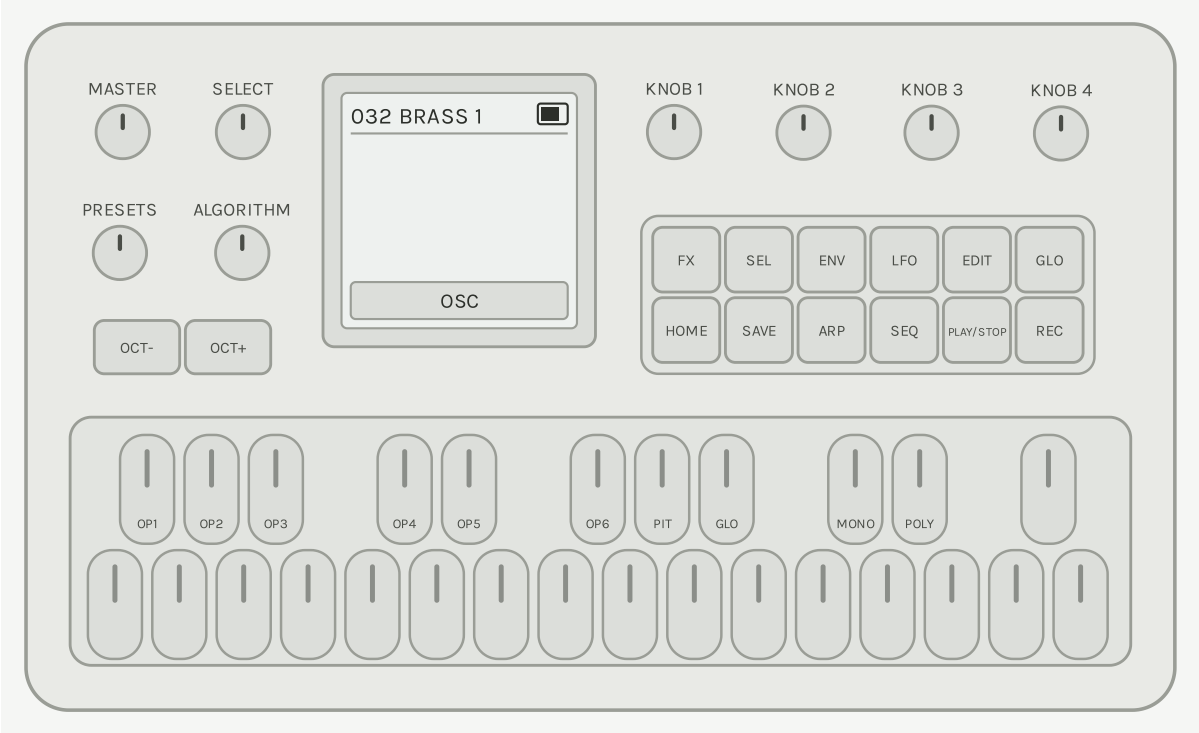
The M-VAVE FM-1

A Worked Tutorial



The front panel, straight down.

by Charlie Wakely

Please note: I have used Claude from Anthropic to help organise, illustrate and construct this guide.

Written for firmware v15.

Released under CC BY-SA 4.0. Copy it, translate it, print it, sell it. Two conditions: say where it came from, and release your version under the same licence. The point is that people learn to use a nice little synthesizer, not that anyone makes money - but if selling copies is what gets it into someone's hands, do that. Copies are available to download at no charge at `github.com/KingParamount/fm1-factory-presets`.

Contents

0. Before we start 3
1. Find a sound you like 7
2. Building a sound from nothing 10
3. The shape of a note 19
4. Modulators 29
5. Algorithms 39
6. Effects 45
7. Movement 52
8. Playing across the keyboard 61
9. The arpeggiator 68
10. The sequencer 72
11. The FM-1 and other things 78

Appendix A - Control changes 82

Chapter 0 - Before we start

Five minutes. Nothing to learn.

The FM-1 is a synthesiser roughly the size of a hardback book, and a surprising amount of it is knobs. Every sound it makes is built out of sine waves - the plainest, smoothest tone there is - by using some of those waves to bend the others out of shape. That turns out to be enough to produce bells, basses, electric pianos, and a great many noises nobody has got round to naming yet.

We'll get to how that works. First, three practical things.

0.1 Turning it on

The power switch is on the back, at the right-hand end, next to the sockets. It's a switch rather than a button, so it stays where you put it and you can always see at a glance whether the instrument is meant to be on.

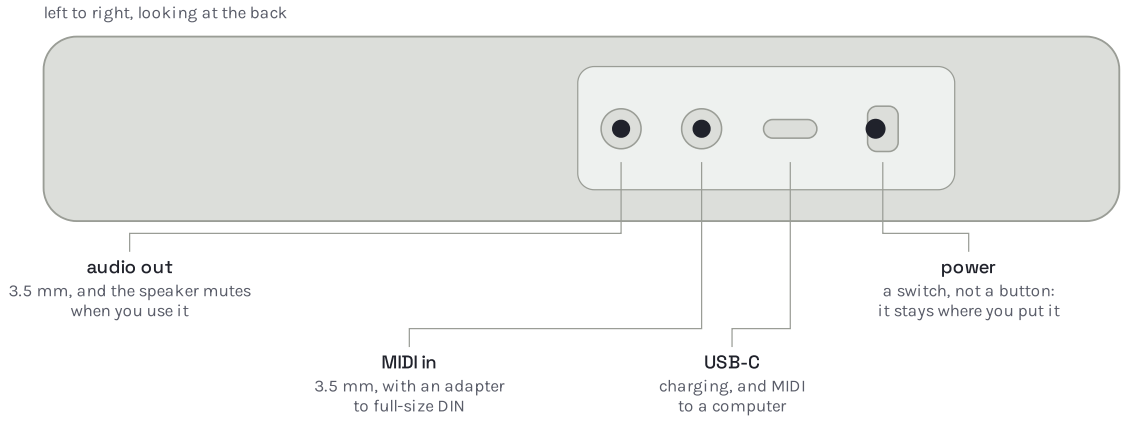


Figure 0.2 - The rear panel, left to right.

0.2 Charging it

Charging happens through the USB-C socket on the back, and a full battery is good for about twelve hours of playing. In practice that means you can forget about it for a week at a time, which is exactly what you want from something you'd like to pick up and play without ceremony.

There's a battery icon in the top-right corner of the screen, and when things get genuinely low the FM-1 puts a full-screen LOW BATTERY message over everything until you plug it in. It's hard to miss, which is the point - it would be a shame to lose a good idea halfway through because the battery ran out quietly.

> ### The keys are a display as well as a keyboard
>
> The keys light from inside, through the words printed on them and through a clear window running down the centre of each one. That is not decoration, and a good deal of what the FM-1 wants to tell you appears there and nowhere else. The screen will not mention any of it.
>
> OP1 to OP6 light to show which operator you are editing, which ones you could switch to, and which ones you have silenced. OCT– and OCT+ light to tell you how far the keyboard has been shifted, by how fast they blink. In the sequencer, a light walks along the lower row showing which step is playing. The light in the HOME button tells you whether Bluetooth is on: flashing while it is searching, solid once it is connected. A long press on HOME switches it off.
>
> None of that is written anywhere on the instrument, and it is easy to spend a while staring at a 1.54 inch screen wondering why it will not tell you something the keys have been announcing all along. Get into the habit of glancing down.

0.3 Hearing it

There's a speaker built in, so the FM-1 makes noise all by itself with nothing plugged into it at all. Put headphones into the 3.5 mm socket on the back and the speaker mutes automatically.

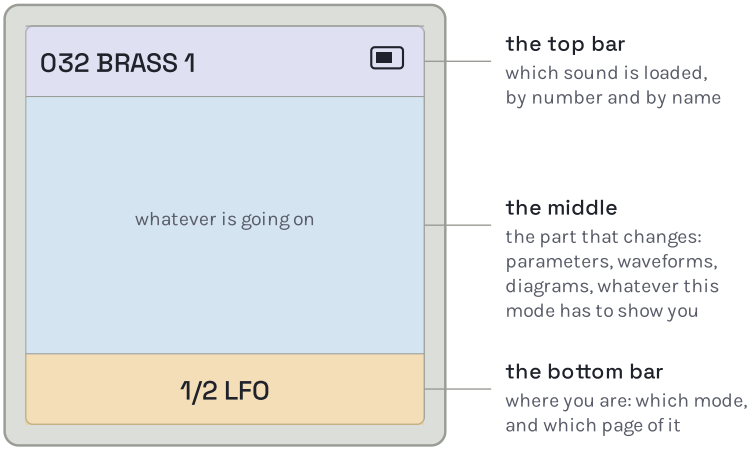
> **Use ordinary headphones**
>
> The manual is specific about this, and it's worth following: use headphones with a standard TRS plug. Other kinds of plug can put noise on the output, and can stop the internal speaker muting the way it should - which leaves you hearing the same notes twice, very slightly out of step with themselves.
>
> A four-pole TRRS headset, the kind with a microphone contact, worked correctly here, speaker muting included. That is one headset on one unit, so treat it as a reprieve rather than a rule.

0.4 The screen

The screen is divided into three strips, and it's the same three strips whatever the FM-1 happens to be doing at the time.

1. **The top bar** shows which sound is loaded, by number and by name - something like `032 BRASS 1`.
2. **The middle** is the part that changes: parameters, waveforms, diagrams, whatever the current mode has to show you.
3. **The bottom bar** tells you where you are - which mode you're in, and which page of it, like `1/2 LFO`.

That bottom bar is going to become the most useful thing on the whole instrument by about chapter two, so it's worth noticing now that it's there.



the same three strips, whatever the FM-1 is doing

Figure 0.3 - The screen, whatever it happens to be showing.

That's chapter zero. Go and turn it on.

Chapter 1 - Find a sound you like

No theory at all. Just listening.

There are a hundred and twenty-eight sounds already loaded into the FM-1, put there by people who know it extremely well, and the only job in this chapter is to find out which of them you like. Nothing here needs understanding. You can come back and read the rest of the document weeks later and this chapter will still have been worth doing.

Volume

The knob marked MASTER is the volume, and it's the only knob on the instrument that stops when it reaches the end - everything else spins round and round forever, which takes a little getting used to. Turning it also brings up a small `Volume` message for about a second so you can see where you've got to.

The keyboard

There are twenty-seven keys, made of soft silicone, and each one lights up as you press it. They're small, and they're not pretending to be a piano - this is an instrument you can put in a rucksack, and something had to give.

Straight out of the box the keyboard covers roughly F3 up to G5, which is about the range of a singing voice. If you want notes above or below that, the two buttons marked OCT− and OCT+ shift the whole keyboard by an octave at a time, up to three octaves in either direction.

How do you know how far you've gone? The two buttons are backlit, and they tell you:

- Both unlit - you're where you started.
- A slow blink - one octave away.
- A fast blink - two octaves.
- Solid on - three, which is as far as it goes.

The button that lights is on the side you've travelled: OCT− if you've gone down, OCT+ if you've gone up.

An octave is a big jump, though. What if you want to move by less than that? Hold down either OCT button and turn the knob marked ALGORITHM. Instead of doing its usual job, it shifts the keyboard one semitone at a time - up to twelve either way, which is an octave the long way round. The screen shows you the number as you go. That knob has another job entirely, which chapter five is about; for now it's just a convenient thing to turn.

And if you ever lose track entirely, pressing both OCT buttons together puts both the octave and the semitone shift back to zero.

Changing the sound

The knob marked PRESETS scrolls through all 128 sounds, numbered 001 to 128. The sound changes the moment you turn it - there's nothing to press to confirm - and if you keep going past 128 it simply wraps round to 001 again.

While you're scrolling, a list appears in the middle of the screen showing the number, the name, and where you are in the sequence, and it fades away about a second after you stop turning.

> **Try this**
>
> Go through all 128 of them. Not carefully - just turn the knob and play two or three notes on each one. It takes about ten minutes, and it's probably the most useful ten minutes you'll spend on this instrument all year.
>
> Write down the numbers of the five you like best. We'll be coming back to them.

The oscilloscope

Press HOME at any point and the screen returns to the oscilloscope: a yellow line on a grey background that moves in time with whatever you're playing.

You don't need it for anything. It is, however, genuinely lovely to watch, and once you start looking at the shapes different sounds make you'll find you can recognise a few of them by eye. HOME is also the button that gets you out of anywhere you've accidentally ended up, so it's a good one to know where to find.

The SELECT knob

The knob beside MASTER is marked SELECT, and there's nothing to do with it yet. Its job is moving between the pages of a screen once you're editing something, which starts in chapter two. There is also a button marked SEL over on the panel, which is a different control doing a different job.

One more knob

There's one more knob worth touching before we stop, marked ALGORITHM. There are thirty-two algorithms, numbered 01 to 32, and turning the knob briefly shows you a small diagram of whichever one you've landed on before it disappears again.

Ignore the diagram for now. We're not going to talk about what it means for another two chapters, and there's a good reason for that - it makes far more sense once you've heard what it does than the other way round.

> ### Try this
>
> Pick one of your five favourite sounds. Hold a note down, and while it's still sounding, turn the ALGORITHM knob slowly all the way round. Listen.
>
> Some of them will sound better than the original. Plenty will sound worse. A few will come out thin, or buzzy, or so quiet you'll wonder whether it's still on. All of that is normal, none of it has broken anything, and by the end you'll have a rough feel for what this knob does long before anyone explains it.

> ### You've changed the sound, but you haven't lost it
>
> The FM-1 only commits a sound to memory when you press the SAVE button, so nothing you do by turning knobs is permanent until you say so. Scroll to a different preset and back again and the original comes back untouched. Every control on the instrument behaves this way, ALGORITHM included.

That's chapter one, and you now know how to make it louder, move the keyboard around, and find a sound you like. Which is, honestly, most of what you need in order to sit down and play the thing. Everything from here on is about making sounds that nobody has made before.

Chapter 2 - Building a sound from nothing

A single sine wave, and what shape it is.

Everything the FM-1 does is made of sine waves - not "based on" or "a bit like", but literally, there is nothing else inside it. There are six sine wave generators, and every one of the 128 sounds you listened to in chapter one is those same six waves arranged in some particular way.

Each of the six is called an operator. It's an odd word for something that just makes a tone, but it's what they've been called since 1983 and it's what everyone who works with these instruments will call them, so it's worth getting used to. On the FM-1 they're labelled OP1 through to OP6.

In this chapter we're going to switch five of them off and look very carefully indeed at the one that's left.

2.1 Getting into the workshop

Press the button marked EDIT and two things happen at once. The screen changes, which you'd expect, and the keyboard stops being a keyboard, which you probably wouldn't.

In edit mode, eight of the keys turn into selector buttons. They're labelled on the key face, below the window - OP1 to OP6, then PIT and GLO, along with MONO and POLY. They blink to show they're available to pick, and whichever one you choose goes solid.

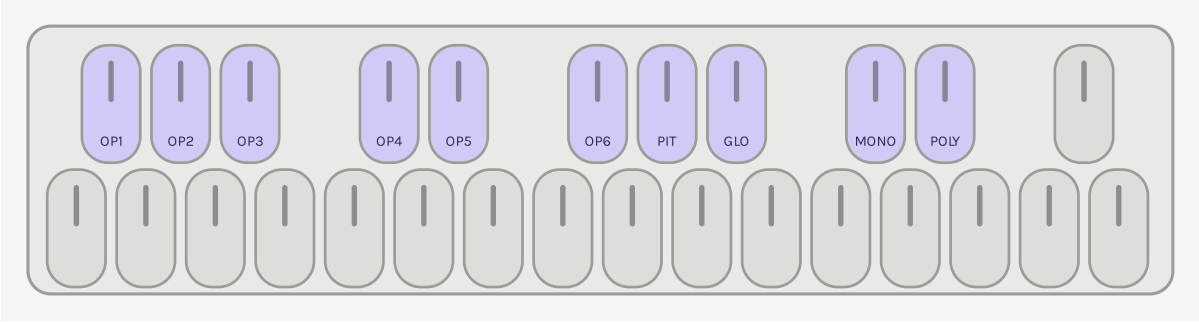


Figure 2.1 - In edit mode the labelled keys stop being notes and become buttons.

Two things are marked GLO, and they are not the same control

One is a button in the top row on the panel, beside EDIT. The other is the key you have just read about, which is only a control at all while EDIT is lit. Pressing them takes you to two different screens, and both screens call themselves GLOBE at the bottom of the display.

This book never writes GLO on its own without saying which. **GLO reached through EDIT is the key. GLO on its own, with EDIT unlit, is the button.** In the route strips from section 2.4 onwards, a control that follows EDIT is always a key on the keyboard, and a control that starts a route is always a button on the panel.

The other labelled keys - OP1 to OP6, PIT, MONO and POLY - have no button of the same name, so there is nothing to confuse them with. GLO is the only label on the instrument that appears twice.

If the keyboard has stopped making any noise

You're still in edit mode. Press EDIT again to come back out. It'll catch you out plenty of times before you start checking that light without thinking about it - and then one day you'll notice you've stopped being caught out at all.

2.2 Starting from nothing

We could take one of the presets apart to see how it works, and that's a perfectly good way to learn - but it's much easier to hear what one thing does when there's only one thing there. So we're going to start from a blank sound instead.

1. Press EDIT.
2. Press the key labelled GLO on the keyboard, not the GLO button on the panel. It stands for global, and it's where the settings live that belong to the whole sound rather than to one operator at a time.
3. Turn SELECT until the bottom bar tells you you're on page 2.
4. Turn KNOB 1 clockwise. That's Init Voice.

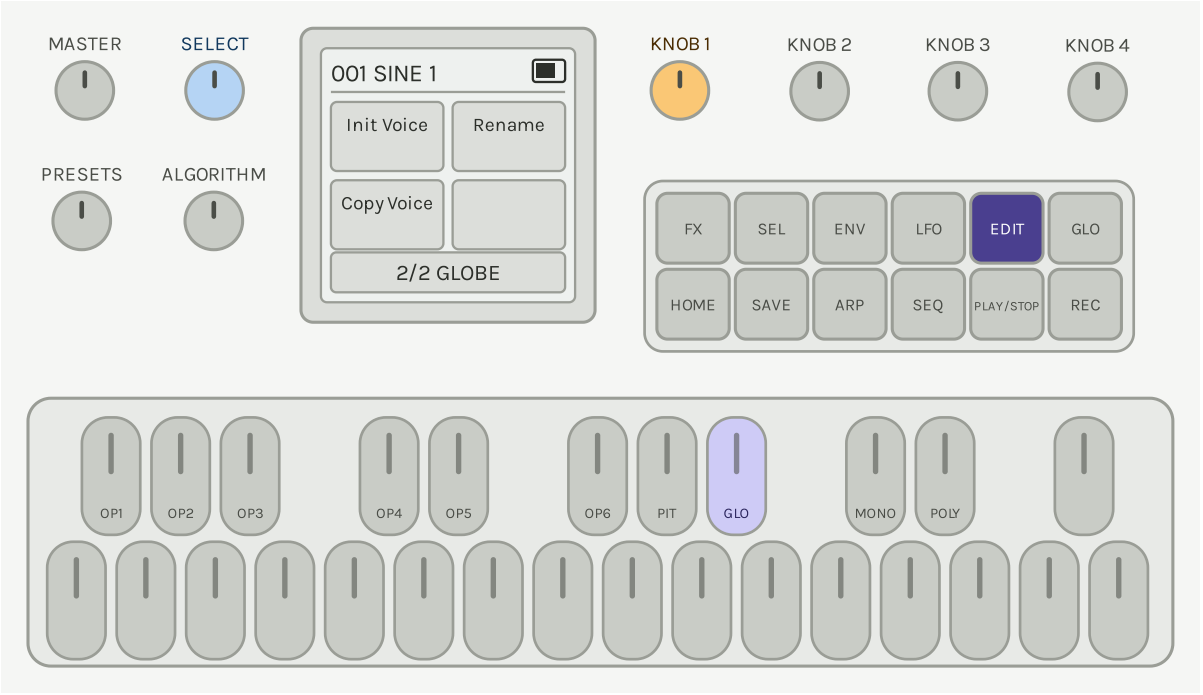


Figure 2.2 - Init Voice, on the patch Globe page.

Every parameter in the sound has now gone back to its factory default. Play a note and see what you've got.

It should be plain, soft and a little dull - not far off the sound of blowing across the top of a bottle. It isn't exciting, and it isn't meant to be. What you're listening to is a single operator producing a pure sine wave, with the other five turned all the way down to nothing.

> **If it sounds like something else entirely**
>
> Bright, buzzy, or completely silent all mean the init didn't take. Go back and do it again - EDIT, then GLO, page 2, KNOB 1 clockwise - and this time check that the screen actually says `Init Voice` before you turn the knob.
>
> Note that Init Voice does not ask whether you are sure, which is unlike almost everything else on the instrument. It simply wipes the sound and says nothing about it. Nothing is lost, though: it only touches what you are working on, not what is stored. Scroll to another preset and back and the original comes home.

2.3 Carriers

The operator you can hear is doing the simplest job an operator can do: making a sound and sending it to the output. An operator doing that particular job is called a carrier.

The other five haven't gone anywhere. They're still sitting there, switched on, set to zero. In the next chapter we're going to turn one of them up - and it won't get louder, it'll get brighter. That single surprising fact is what this whole instrument is built on, and it's the reason FM synthesis is worth the effort of learning.

For now, though: one carrier, one sine wave, nothing else running.

2.4 How this instrument is steered

Before we change anything, it's worth setting out the grammar, because every instruction in the rest of this document is made of the same few parts and never any more than that:

1. Which button - the area you're working in. Always first, always a button on the panel.
2. Which key - where the button has turned the keyboard into a row of selectors. Only ever appears after a button, never instead of one.
3. Which page - turn SELECT until the bottom bar agrees with you.
4. Which knob - one of KNOB 1 to KNOB 4.

Not every instruction needs all four. From here on they appear together as a strip, like this:

EDIT OP1 page 1/6 KNOB 4

Read it from left to right, do what it says, and you'll be in the right place. The dark pills are buttons on the panel and the pale ones are keys on the keyboard, which is the difference between the two things marked GLO: GLO is the button, and EDIT GLO is the key.

2.5 The four knobs and the four boxes

KNOB 1 to KNOB 4 don't have fixed jobs. What each one does depends entirely on which page you happen to be looking at, and the bottom bar always tells you what they're controlling at that moment.

A page never holds more than four things. They appear as four boxes on the screen, and the knobs map onto them in reading order: KNOB 1 is the top-left box, KNOB 2 the top-right, KNOB 3 the bottom-left, and KNOB 4 the bottom-right.

The knobs sit in a row and the boxes sit in a square, so the mapping is one you learn rather than one you can see. Look at the diagram once, properly, and it tends to stick.

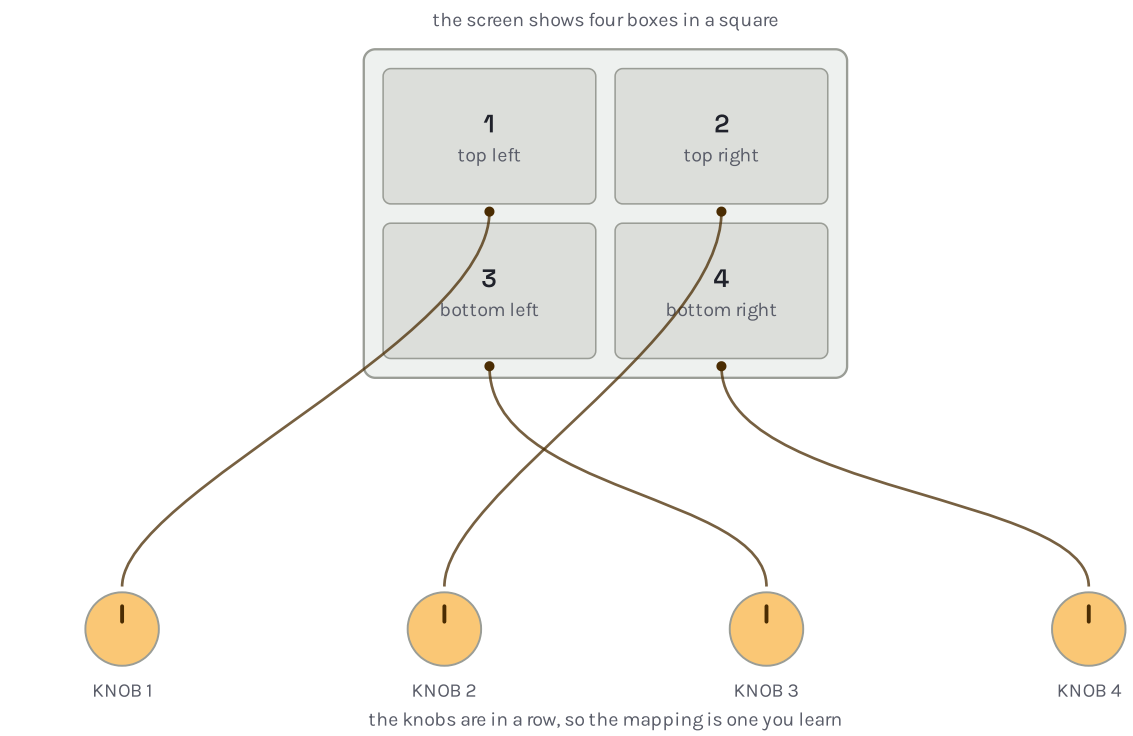


Figure 2.3 - Four knobs in a row, four boxes in a square. Worth one careful look.

The FM-1 remembers where you left off

Go back to an operator you were editing earlier and it returns you to whichever page you were last looking at, rather than to page 1. That's helpful when you're working, and confusing when you're following instructions.

So get into this habit straight away: **press the button, then turn SELECT anticlockwise until the bottom bar says page 1.** Then count forwards from there. Every instruction in this document assumes you've done exactly that.

2.6 Making the one note louder and quieter

Time to change something. Press EDIT, press OP1, and get yourself to page 1 - the bottom bar will read `1/6`. This page is called Tuning & Level, and it holds four parameters: Detune, FreqCoarse, FreqFine and Out Level.

EDIT OP1 page 1/6 - Detune, FreqCoarse, FreqFine and Out Level, on knobs 1 to 4.

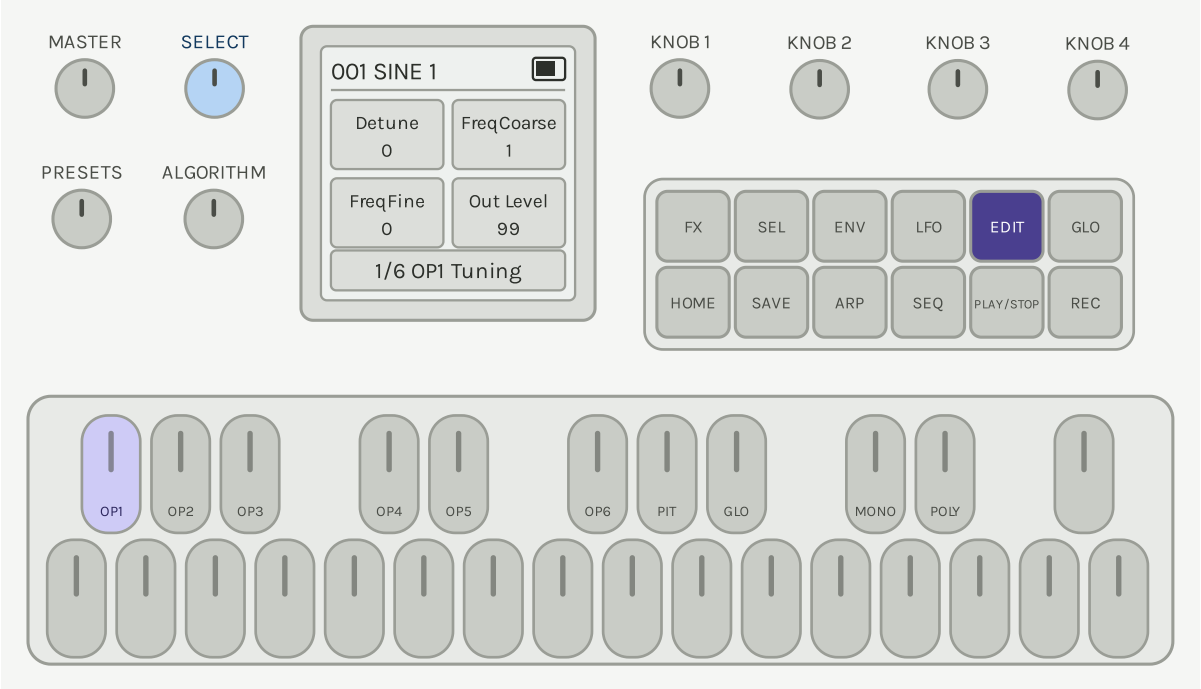


Figure 2.4 - OP1, page 1.

> **Out Level** (KNOB 4, 0-99) - how much this operator puts out. Hold a note down and turn it all the way to zero. Silence. Bring it back to 99 and the sound returns.

> **FreqCoarse** (KNOB 2, 0-31) - what pitch this operator runs at, relative to the key you pressed. Have a turn of it if you're curious. On a carrier all it can do is move the note up in jumps, which is mildly interesting at best.

> **FreqFine** (KNOB 3, 0-99) - the same, in smaller steps.

> **Detune** (KNOB 1, -7 to +7) - a very small amount either side of in-tune.

So Out Level is the volume of this operator - which might seem like a slightly redundant thing to have, given there's already a volume knob on the front panel.

> **Worth remembering**
>
> It isn't really volume. It only looks like volume today, because OP1 is a carrier and there's nothing else switched on for it to interact with.
>
> Turn the very same knob on an operator that isn't a carrier and it controls how bright the sound is instead - same page, same position, entirely different result. Which one you get depends on what job that operator has been given.
>
> That's chapter four.

The three tuning controls become some of the most important on the instrument in chapter four, once the operator you're retuning is one you can't directly hear.

Put anything you moved back where it was before you go on.

2.7 Keeping it

You've made a sound. It isn't a very good sound, but it's yours, and everything in chapter three gets built on top of it - so it's worth saving properly rather than starting from scratch next time.

> **What is on each of the two GLOBE screens**
>
> Both screens call themselves GLOBE at the bottom of the display, both have two pages, and the manual lists their contents in a single table as though they were one thing. They are not one thing.
>
> GLO is the instrument. Page 1 is `Midi Chn`, `Effect Chn`, `Pitch Up` and `Pitch Dn`. Page 2 is `Reset Patches`, `Velocity`, `Glide` and `Glide Val`.
>
> EDIT GLO is the sound you are working on. Page 1 is Algorithm, Feedback, Osc Sync and Transpose. Page 2 is Init Voice, Rename and Copy Voice, on three knobs rather than four.
>
> Turning a knob on the wrong one is a confusing way to spend ten minutes, so the route is written out in full wherever this book sends you to either.

Give it a name first. Press EDIT, press GLO, turn SELECT to page 2, and turn KNOB 2 clockwise to open the rename panel. The keys OP1 to OP5 choose what kind of character you're entering, KNOB 3 picks the character itself, and SELECT confirms it. If you change your

mind, turning KNOB 2 anticlockwise backs out without altering anything.

Call it something you'll recognise in three weeks' time. `SINE 1` does the job.

Then press SAVE. The manual says to press and hold it, but a tap does the same thing. SAVE lights while it's writing, the screen shows `All Data Saved`, and that's that - it's in flash memory, and it'll still be there after the FM-1 has been switched off and on again.

Saving writes over whichever preset slot you're on. If that makes you nervous, it mostly shouldn't. There is a reset, and it lives on the instrument Globe page, which is the GLO button pressed on its own.

GLO page 2/2 - Reset Patches, Velocity, Glide and Glide Val, on knobs 1 to 4.

1. Press GLO on its own, turn SELECT to page 2, and turn KNOB 1 clockwise. Not EDIT and then GLO. On that page KNOB 1 is Init Voice, and it would wipe the sound you have just saved.
2. A panel appears. Nothing has happened yet.
3. Knobs 1 to 4 choose which bank to reset. The 128 presets are divided into four: A is 1 to 32, B is 33 to 64, C is 65 to 96, D is 97 to 128.
4. Turn SELECT clockwise to go ahead, or anticlockwise to change your mind.
5. The screen says `Reset Success` when it's done.

So for everything this chapter asks you to do, the presets are not precious. Overwrite them, experiment in them, fill them with rubbish, and the reset puts them back.

> **What the reset does not do**
>
> It restores the chosen bank to whatever was last imported into it. If nothing has ever been imported there - which is the case on a new instrument - that is the factory content, and everything comes back.
>
> Import a bank of voices into it, though, and the factory sounds in that bank are gone for good. The reset will hand you the imported bank instead. The FM-1 will not send its presets out to a computer either, so there is no way to back them up first.
>
> They can be restored, but not from the instrument: the factory set is available as a download to send back in, and chapter eleven has the details. It goes into all of this before it asks you to import anything.

Spare copies to work in

One last thing before chapter three. Make two more copies of `SINE 1` in other slots, because chapter three renames two of them.

1. Select a different preset. `002` will do.
2. EDIT GLO page 2/2, then Copy Voice.
3. Pick `SINE 1` with KNOB 4 and confirm with SELECT.
4. Press SAVE.

Do that again into a third slot, so you have three copies of `SINE 1` to work in.

> ### What you have at the end of this chapter
>
> A saved sound called `SINE 1`: one operator, one sine wave, nothing else running at all. Three slots hold a copy of it.
>
> It's the plainest noise the FM-1 is capable of making, and it's also the only sensible place to start from. You'll find yourself coming back to it every time you want to build something new.

What's next

There's one obvious thing missing from this sound. It starts the instant you press a key and stops the instant you let go, which is not how any real instrument behaves. A piano note begins hard and fades away. A violin swells up gradually. A bell goes on ringing long after it's been struck, whatever you do with your hands.

That shape - how loud a note is at each moment between pressing the key and silence - is what chapter three is about. It lives on pages 2 and 3 of the very same operator you've just been editing, and the way the FM-1 handles it is a little unusual, so it gets a chapter of its own.

Chapter 3 - The shape of a note

A note is a shape, and you draw it with four corners.

Load up `SINE 1` from the end of the last chapter and play a note. Hold it. Let go.

Whatever else is wrong with that sound, the biggest problem is that it arrives at full volume the instant you touch the key and vanishes the instant you release it. Nothing in the physical world behaves like that. Strike a piano key and the note is loudest immediately and then dies away, however long you keep your finger down. Draw a bow across a violin and the note creeps up from nothing. Hit a bell and it goes on ringing for ten seconds no matter what you do.

That difference - how loud a note is at each moment between the key going down and silence returning - is called the envelope, and it does more to tell your ear what kind of instrument it's listening to than almost anything else.

3.1 Four corners

Most synthesisers give you four controls for this and call them attack, decay, sustain and release. The FM-1 does something rather more flexible, and it's worth understanding on its own terms rather than trying to squash it into the usual four.

Think of it as drawing a shape. You get to place four corners, and you get to say how quickly the sound travels between them.

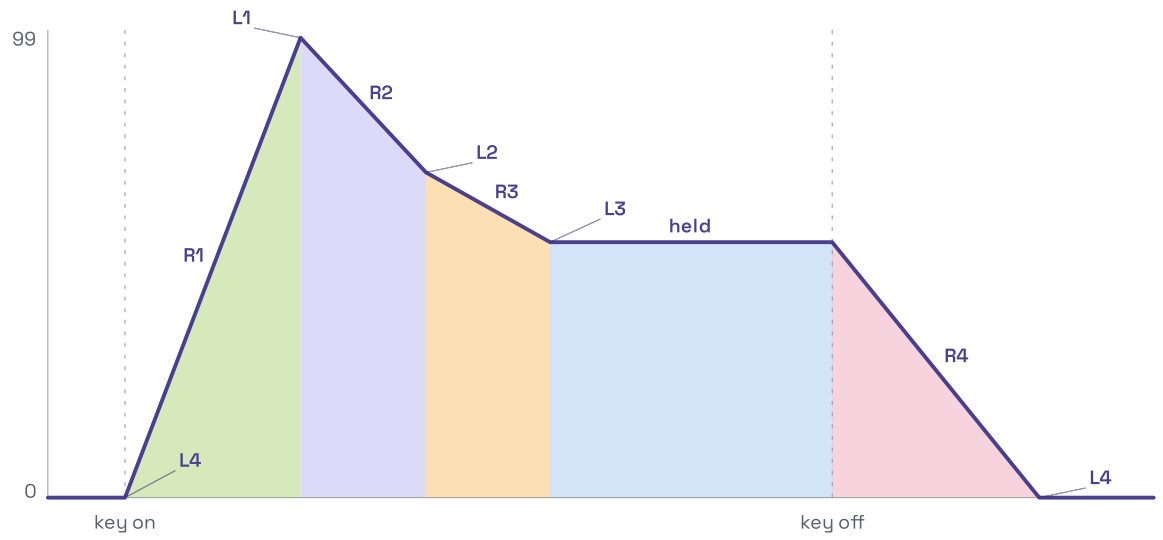


Figure 3.1 - Four corners and four slopes. Every FM-1 sound is one of these.

Reading it from the left: before you press anything, the sound sits at L4. Press a key and it sets off at rate R1 towards L1. When it gets there it turns and heads at rate R2 towards L2, then at rate R3 towards L3. And there it stays, for as long as you keep the key down - a second, a minute, all afternoon.

Let go of the key and it travels at rate R4 back down to L4, where it started.

Four places to be, four speeds for getting between them. That's the whole system, and every sound the FM-1 has ever made uses it.

3.2 Where to find it

The levels and the rates live on two separate pages of whichever operator you're editing.

3.2.1 The rates

EDIT OP1 page 2/6 - Rate 1 to Rate 4, on knobs 1 to 4.

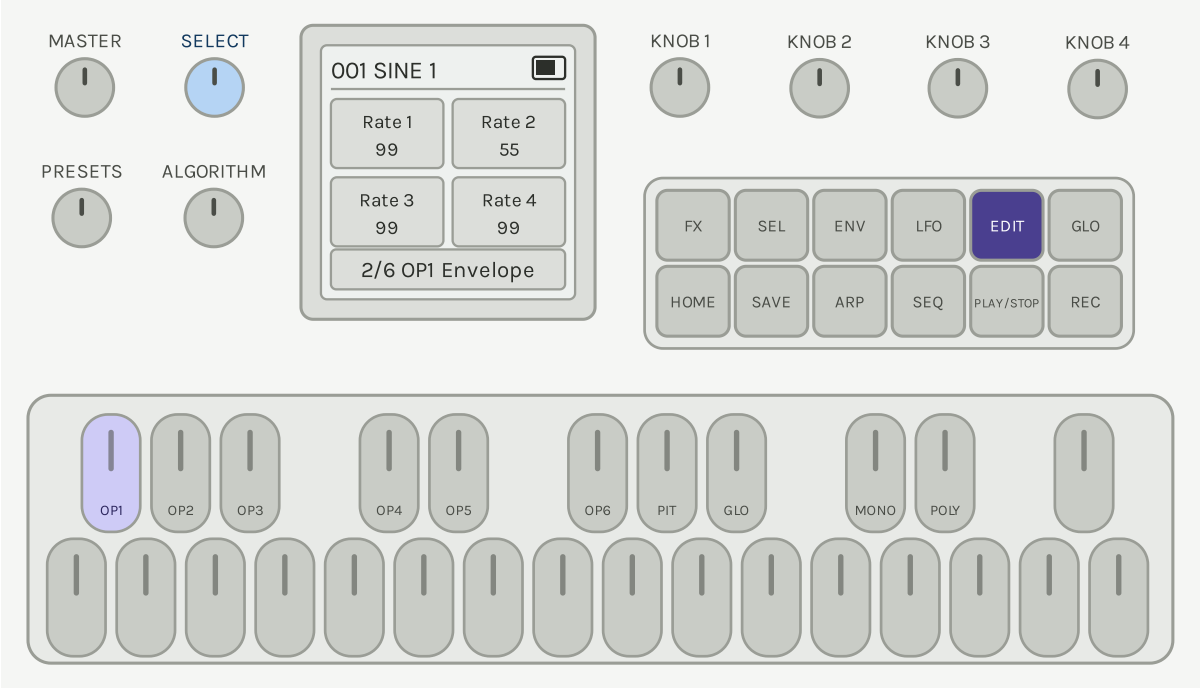


Figure 3.2 - The rate page.

> **R1** (KNOB 1, 0-99) - how fast the sound travels from where it starts up to L1.

> **R2** (KNOB 2, 0-99) - from L1 to L2.

> **R3** (KNOB 3, 0-99) - from L2 to L3, where it settles.

> **R4** (KNOB 4, 0-99) - after you let go: from wherever it was back down to L4.

3.2.2 The levels

EDIT OP1 page 3/6 - Level 1 to Level 4, on knobs 1 to 4.

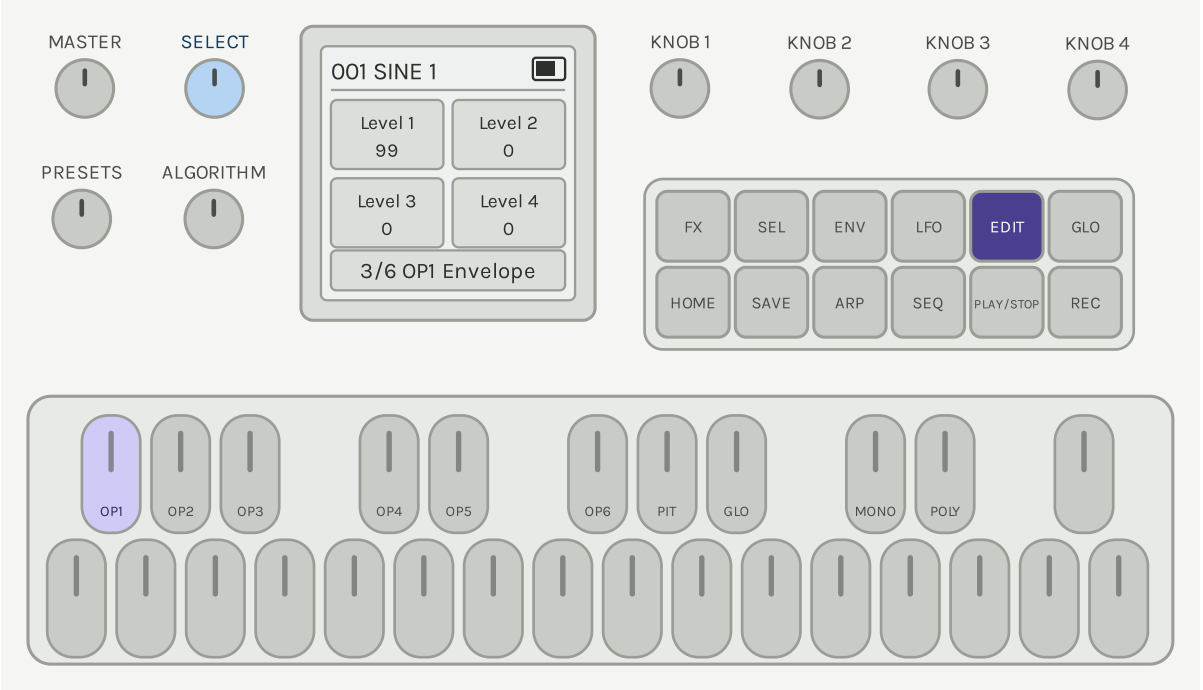


Figure 3.3 - The level page.

> **L1** (KNOB 1, 0-99) - the first corner, which the note heads for as soon as you press the key.

> **L2** (KNOB 2, 0-99) - the second corner.

> **L3** (KNOB 3, 0-99) - where it settles and stays while you hold the key down. Section 3.4 is about this one.

> **L4** (KNOB 4, 0-99) - where it sits before you play, and where it returns to afterwards. Section 3.5 is about why that's strange.

Two pages, eight numbers, one shape.

> **Remember the reset habit**
>
> The FM-1 puts you back on whichever page you last used for that operator. Press OP1, wind SELECT all the way back to page 1, then count forward. Two clicks gets you to the rates, three to the levels.

3.3 The bit that trips everyone up

Here's the thing worth getting straight before you touch anything, because it catches out people who have been doing this for years.

The rate numbers are not durations. They're speeds. A high rate means the sound moves quickly, which means that part of the note is *shorter*.

So turning R1 up makes the attack snappier, not longer. Turning R4 down makes the note hang around after you release the key. It's the opposite of what the word "rate" makes most people expect the first time.

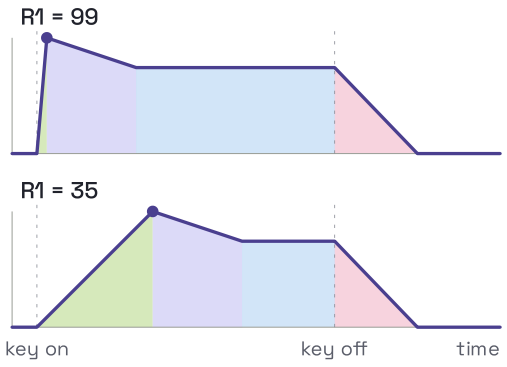


Figure 3.4 - The same levels twice. Only R1 differs, and the note is a different length.

> **Try this**
>
> With `SINE 1` loaded, go to OP1's rate page and turn R1 down to somewhere around 45. Play a note and hold it.
>
> Instead of arriving, the sound now creeps in. Turn R1 back up to 99 and it snaps in again. That one knob is the difference between a bowed instrument and a struck one.

3.4 Where the note sits while you hold it

L3 is the level the sound settles at once it has finished travelling - the level you hear for as long as your finger stays down. It's the one corner whose duration you don't set, because you set it yourself with your hand.

Set L3 to 99 and a held note goes on at full volume indefinitely, like an organ. Set L3 to 0 and the note dies away to nothing whether you hold the key or not, like a piano or a bell.

That single choice - does the sound sustain, or does it decay - accounts for most of the difference between the instrument families you already know by ear.

3.5 The strange one

L4 is where the sound sits before you press anything, and also where it returns to after you let go. It's the same corner doing both jobs.

Which raises a question worth sitting with for a moment: what happens if L4 isn't zero?

The answer is that the sound never actually stops. It sits at whatever level you've set, quietly, all the time, whether you're playing or not - a drone under everything you do.

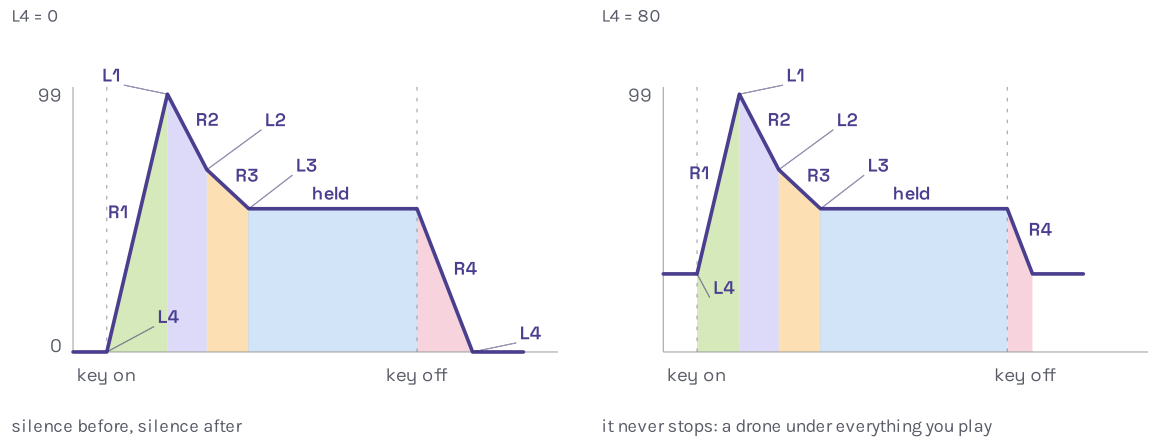


Figure 3.5 - L4 at zero, and L4 raised. On the right the sound never stops.

Try this, then put it back

Set L4 to about 80 and play nothing at all. There it is - a permanent hum.

This is usually a mistake, and when a sound you're building won't shut up, L4 is the first place to look. But it's also a real effect that people use on purpose, so it's worth having heard it deliberately once rather than only ever meeting it by accident.

Put L4 back to 0 before you go on. That alone will not stop the noise. Switch to another preset and then back to `SINE 1`, and it goes quiet.

3.6 Letting go early

One more thing, and it explains a lot of behaviour that otherwise looks like the instrument ignoring you.

Releasing a key doesn't wait for anything. Whatever the note was doing at that instant - still climbing towards L1, halfway down to L2 - it stops and heads straight for L4 at rate R4.

So a sound with a slow attack will never reach full volume if you play it in short stabs. That isn't a fault, it's the whole character of the thing, and it's why slow-attack sounds want to be played with long held chords.

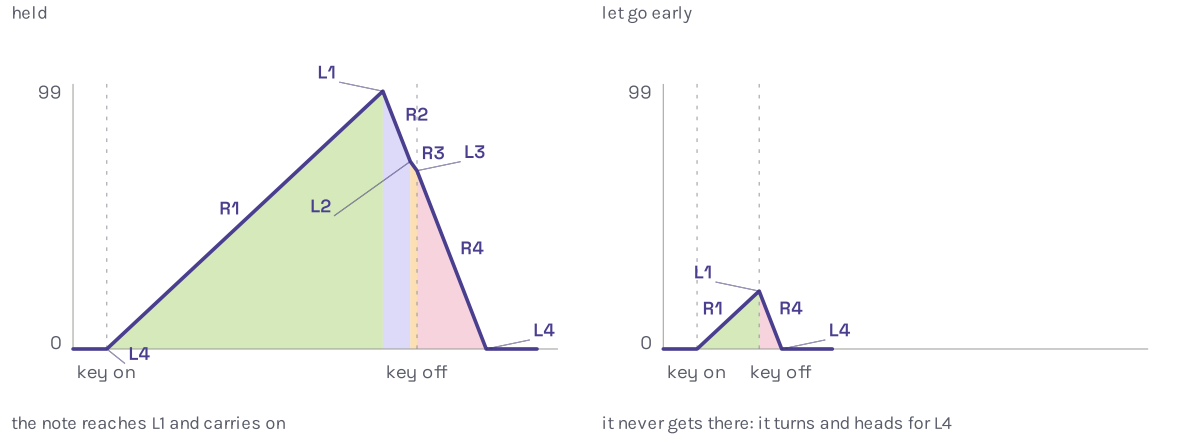


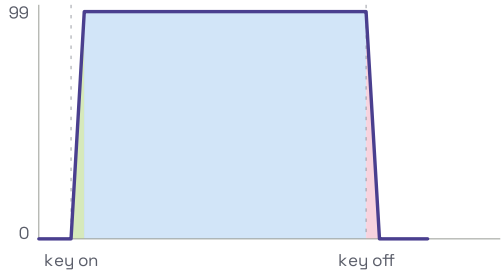
Figure 3.6 - Let go partway up and the note turns for L4 without ever arriving.

3.7 Four shapes worth knowing

Here are four envelopes that between them cover most of what you'll want. Type them in on OP1 and listen. Every one of these numbers is a starting point rather than a rule - nudge them until they sound right to you.

Organ

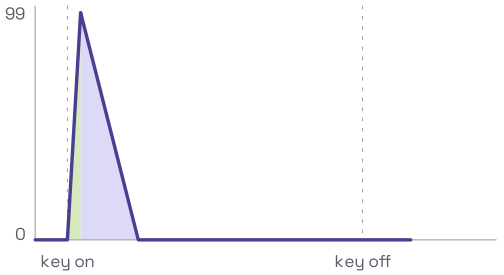
Instant on, full volume for as long as you hold it, instant off.



rates	R1 99	R2 99	R3 99	R4 99
levels	L1 99	L2 99	L3 99	L4 0

Pluck

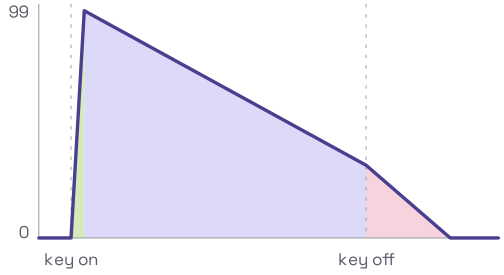
Arrives at once and falls away to nothing, whether you keep holding or not.



rates	R1 99	R2 55	R3 99	R4 99
levels	L1 99	L2 0	L3 0	L4 0

Bell

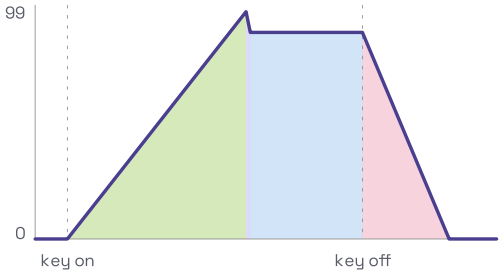
The same fall, far more gradual - so gradual that letting go interrupts it.



rates	R1 99	R2 25	R3 99	R4 30
levels	L1 99	L2 0	L3 0	L4 0

Swell

Creeps up from silence, settles just below the top, fades gently when released.



rates	R1 35	R2 60	R3 60	R4 45
levels	L1 99	L2 90	L3 90	L4 0

The colours are the same ones from diagram 3.1, so each number above is tinted to match the region of the shape it controls. Change a green number and the green part of the picture moves. That correspondence is worth more than anything else on this page.

Look at what the bell does

Every shape here is drawn with the key held for the same length of time and released at the dashed line. Three of them have finished travelling by then and are sitting still. The bell hasn't - it's still on its way down when you let go, which is why its pink release region is the only one you can see.

That is also why R4 does nothing at all on the pluck. By the time you release the key the sound is already at zero, so there is nowhere left for it to travel.

Try this

Take the pluck and change one number at a time. Bring R2 down to 20 - now it's a bell. Bring L2 up to 70 - now it's an organ with a bump at the start. Bring R4 down to 20 - now every note trails off after you let go, and playing fast makes them pile up on top of each other.

You aren't looking for the right answer here. You're building up a feel for which knob does what, which is worth more than any table.

Keeping two of them

Build the pluck and swell sounds in the spare `SINE 1` copies from chapter two, so you are not saving over `SINE 1` twice in a row. Then save them as separate sounds - EDIT, then GLO, page 2, KNOB 2 to rename, then press SAVE - and call them something like `PLUCK 1` and `SWELL 1`.

What you have at the end of this chapter

Three saved sounds: `SINE 1`, `PLUCK 1` and `SWELL 1`. All of them one sine wave, all of them plain, but now with completely different characters - because the shape of a note carries more information than the note itself.

If you fancy something odder

The key marked PIT holds a second envelope, built exactly the same way - four levels on one page and four rates on the other. The difference is where it goes: instead of controlling how loud the sound is, it controls what pitch it's at.

One thing to know before you touch it: on the pitch envelope, 50 means "no change". Above 50 bends the note up, below 50 bends it down. So a level of 0 isn't silence, it's a long way flat.

The other thing to know is that the numbers are wildly uneven. Going from 50 to 70 buys you about a fifth. The next ten steps buy another third or so, the ten after that most of an octave, and the last few run away entirely. Somewhere around 85 it stops being a pitch control and becomes a special effect.

So the useful range is roughly 50 to 80, and it is worth knowing that before you spin the knob, find yourself several octaves out, and conclude the thing is broken.

Try L4 at 99 with everything else at 50, and R1 somewhere around 75 - a note that swoops down into place from above. Setting R1 to around 40 produces the sound used in every computer game until about 1985.

Chapter 4 - Modulators

An operator can change another operator instead of being heard.

Everything so far has used one operator out of six. This chapter turns on a second one, and the result is not what you'd expect - which is the whole reason this instrument exists.

Load `SINE 1` and let's find out.

> **The most useful button nobody mentions**
>
> In EDIT mode, press and hold any OP key and that operator goes silent. Its light goes out so you can see which ones you have muted. Press and hold again to bring it back, and changing preset clears the lot.
>
> This chapter and the next spend their length arguing that some operators are heard directly and some are only heard through others. You do not have to take that on trust. Mute them one at a time and listen: silence one and the sound disappears, silence another and the tone changes character without going away. The first kind is a carrier. The second is a modulator.
>
> It only works from EDIT, and it is never saved with the patch. It is a way of listening rather than a way of building, which is exactly what you want here.

4.1 Turning up the one you can’t hear

OP1 is the one you've been listening to. OP2 has been sitting there at zero the entire time.

EDIT OP2 page 1/6 KNOB 4 - Out Level.

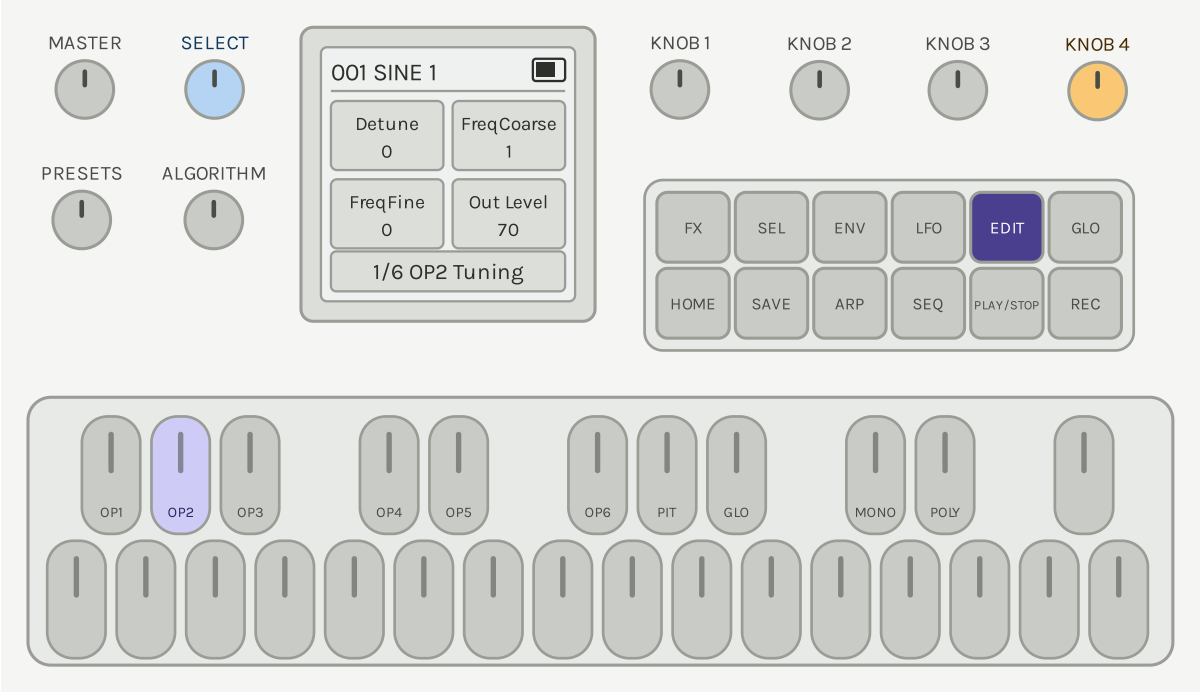


Figure 4.1 - OP2, output level.

Repeatedly play a note, and slowly bring that from 0 up to about 70.

The sound doesn't get louder. It gets brighter - harder, buzzier, more like a real instrument and less like a test tone. Keep going to 99 and it turns harsh and metallic.

> ### This is the thing
> Same knob, same page, same position as the one that controlled volume in chapter two. Completely different result.
>
> The difference is the job OP2 has been given. It isn't connected to the output at all - it's connected to OP1. An operator wired to the output is a carrier and you hear it. An operator wired to another operator is a modulator, and you never hear it directly; you only hear what it does to the thing underneath.

4.2 Why brightness, though?

Here's what's actually happening, and it's worth a minute because everything else in FM synthesis follows from it.

A modulator pushes the carrier's pitch up and down. That's all it does. Turn the modulator's level up and it pushes harder.

Now, if something pushes a note's pitch up and down a few times a second, you hear it as vibrato - a wobble. Speed that wobble up and at some point it stops sounding like a wobble at all. Your ear can't track it any more, so it does something else with the information: it hears it as a change in tone.

A modulator is doing vibrato several hundred times a second. Far too fast to hear as movement, so you hear it as brightness instead. And the harder it pushes, the brighter the result.

That's the trick. That's the whole of it. Everything from here is just deciding what to push with, and how hard, and when.

4.3 What note is it pushing at?

The modulator has its own pitch, and this turns out to matter far more than the modulator's own pitch ever would if you could hear it. Everything that sets it lives on the page you're already on.

EDIT OP2 page 1/6 - Detune, FreqCoarse, FreqFine and Out Level, on knobs 1 to 4.

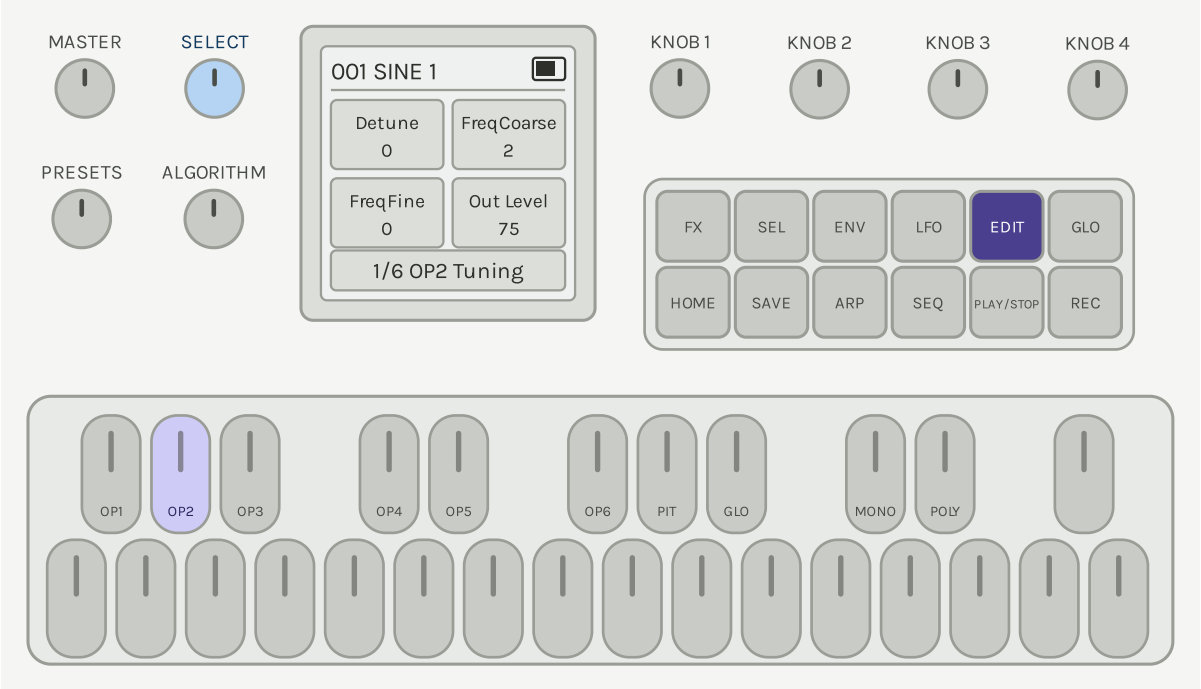


Figure 4.2 - OP2, tuning. All four controls live on this page.

> **FreqCoarse** (KNOB 2, 0-31) - a multiplier rather than a pitch. Set it to 1 and the modulator runs at the same frequency as the note you played. Set it to 2 and it runs twice as fast, at 3 three times as fast, and so on.

> **FreqFine** (KNOB 3, 0-99) - fills in the gaps between the whole numbers.

> **Detune** (KNOB 1, -7 to +7) - a very small amount of not-quite-right, applied deliberately. Section 4.3.3 is about why you'd want that.

> **Out Level** (KNOB 4, 0-99) - how hard it pushes, which you have already met.

4.3.1 Whole numbers

> **Try this**
>
> With OP2's level at about 70, hold a note and step FreqCoarse through 1, 2, 3, 4, 5 and so on up to about 12. Listen to each one for a few seconds before moving on.
>
> They all sound different, and they all still sound like a musical note - thin, hollow, reedy, nasal, brassy. What you're hearing is the same instrument playing the same pitch with a different tone each time.

The reason they stay musical is that whole-number multiples of a note are exactly the frequencies that occur naturally in real instruments. Doubling gives you the octave above; tripling gives you a fifth above that. Stack them up in different amounts and you get flutes, clarinets, oboes, trumpets - everything that vibrates in a well-behaved way.

4.3.2 Breaking it on purpose

So what happens if the multiplier isn't a whole number?

> **Try this**
>
> Set FreqCoarse to 3, then wind FreqFine up towards the middle of its range so you're somewhere around three and a half times the note's frequency. Play something.
>
> It stops sounding like a note and starts sounding like an object. Bells, gongs, metal bars, glass. Suddenly there's a percussion section in here that wasn't there a moment ago.

The frequencies you've created are no longer whole-number multiples, so they don't line up with anything your ear expects from a vibrating string or column of air. It hears metal instead. This is why FM synthesis was famous for bells and electric pianos long before it was famous for anything else - it's extremely good at the sounds that other kinds of synthesis find awkward.

4.3.3 Close, but not quite

There's a third option between those two, and it's the most useful of the lot.

> **Try this**
>
> Put FreqCoarse back to a whole number - 2 is a good one - and then move Detune just a tiny way off zero. Not far. A few clicks.
>
> Play a long held note and wait.

The sound moves. It shimmers, or beats slowly, or seems to breathe - because two frequencies that are nearly but not quite in step drift gradually in and out of alignment with each other.

This is the difference between a sound that's obviously synthetic and one that seems alive. Real instruments are never perfectly in tune with themselves; strings stretch, air moves, wood warps. That is what Detune is for.

4.4 When the brightness changes

Now for the part that turns all this into actual instruments.

The modulator has an envelope of its own - pages 2 and 3 of OP2, exactly as you learned in chapter three - and it works in precisely the same way. But because OP2's level controls brightness rather than volume, its envelope controls how the brightness changes over time.

Think about what that means for a moment. Think about hitting a piano key hard: the sound is bright at the very start and then mellows, even while the note is still ringing. That's not the volume changing, it's the tone changing. Almost every acoustic instrument does this.

> **Build an electric piano**
>
> Start from `SINE 1`, then:
>
> **OP1** - the pluck envelope from chapter three, with R2 opened out a little.
>
> rates `R1 99` `R2 40` `R3 99` `R4 99` levels `L1 99` `L2 0` `L3 0` `L4 0`
>
> **OP2** - level about 75, FreqCoarse 2, and its own envelope set to decay much faster than OP1's.
>
> rates `R1 99` `R2 50` `R3 99` `R4 99` levels `L1 99` `L2 0` `L3 0` `L4 0`

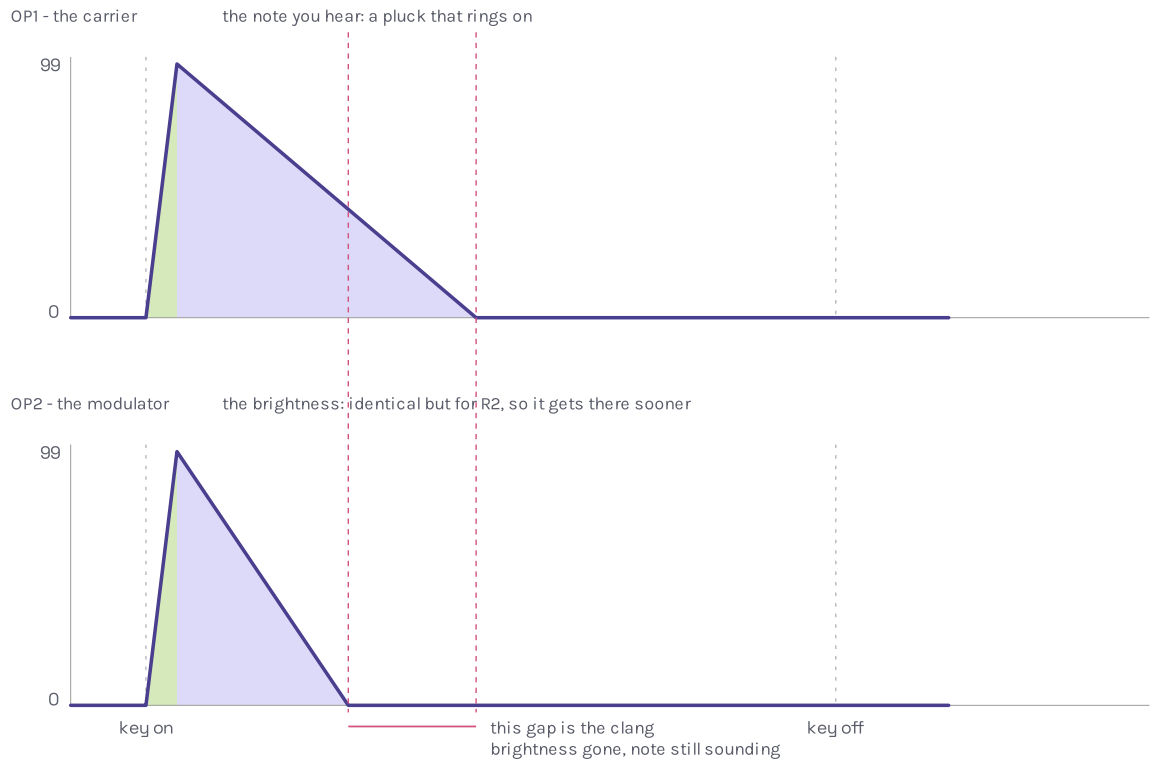


Figure 4.3 - The electric piano. One number differs, and it is the instrument.

Play it. The note starts bright and clangy and settles almost immediately into something soft and round underneath, and it should be recognisably an electric piano.

The two shapes are almost the same. The only number that differs is R2, and the whole instrument lives in the gap between them: the note is still sounding there, but the brightness has already gone. That gap is the clang.

Then go back to OP2's R2 and slow it down. The clang lasts longer. Speed it up and it becomes a click. That single number is the difference between a whole family of instruments.

> **Two envelopes, two jobs**
>
> The carrier's envelope shapes how loud the note is. The modulator's envelope shapes how bright it is. They're completely independent, and using them against each other - bright and quiet, or dull and loud - is where most good FM sounds come from.

4.5 Fixed instead of ratio

One option worth knowing about before we move on.

EDIT OP2 page 4/6 KNOB 3 - OscMode, either Ratio or Fixed.

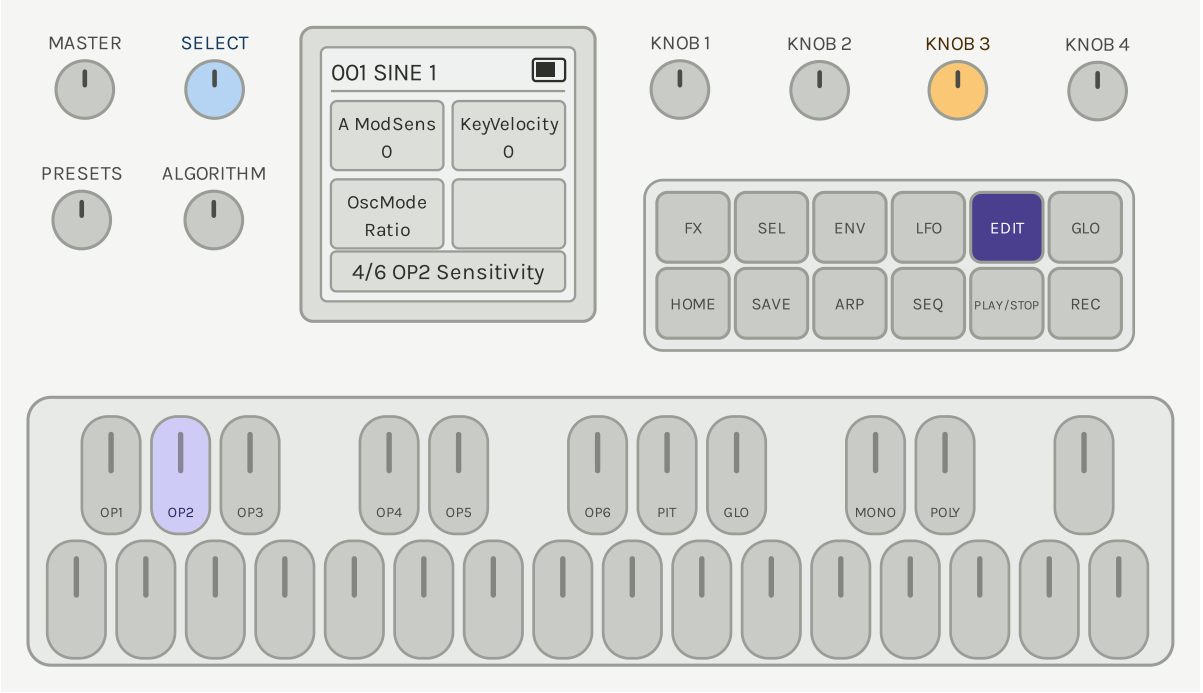


Figure 4.4 - OP2, oscillator mode.

Everything so far has been Ratio: the modulator's frequency follows the note you play, so the sound has the same character wherever you play it on the keyboard. Switch to Fixed and the modulator stays put at one frequency no matter what you press.

That sounds like a mistake, and for most instruments it would be - but it's exactly right for anything where part of the sound doesn't move with pitch. The thump of a bass drum. Breath noise on a flute. The knock of a hammer. Real instruments are full of components that stay the same however high or low you play, and Fixed is how you get them.

4.6 A second modulator

Everything so far has used two operators out of six. Adding a third is where it starts to get properly interesting, and it needs a different arrangement.

EDIT GLO page 1/2 KNOB 1 - Algorithm, 1 to 32. Set it to 16.

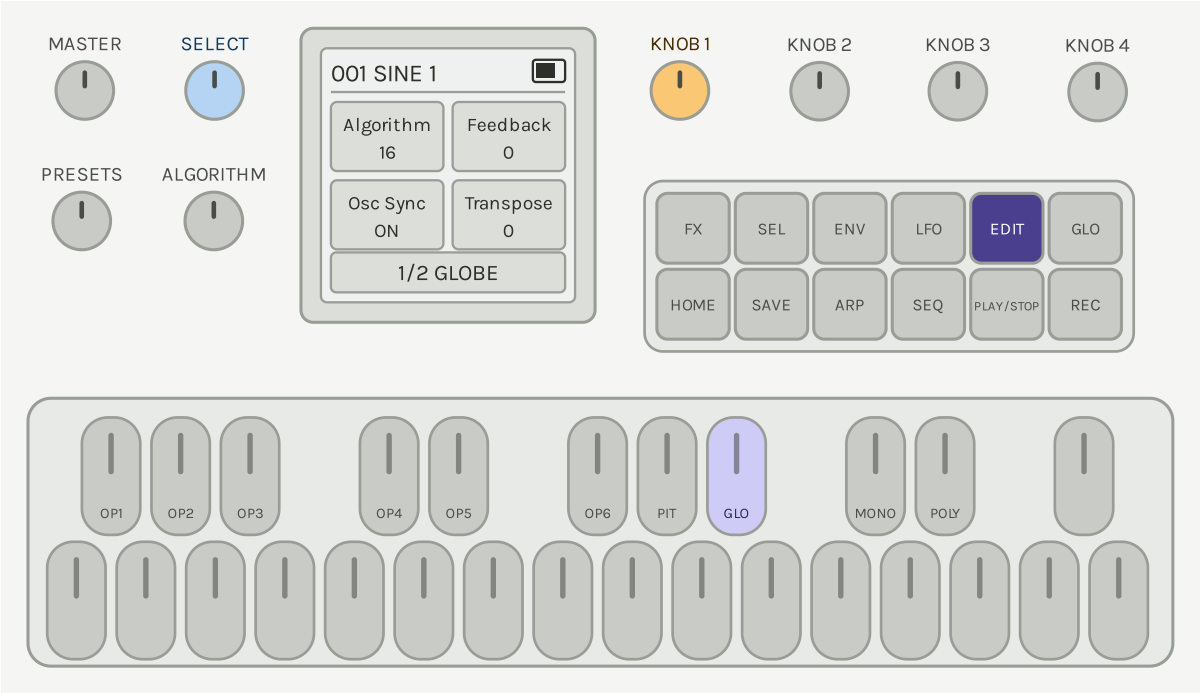


Figure 4.5 - The patch Globe page, algorithm.

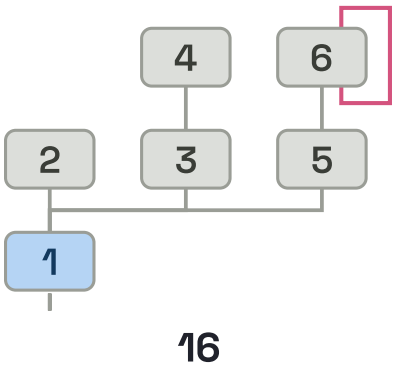


Figure 4.6 - Algorithm 16.

Algorithm 16 has one carrier, OP1, with OP2, OP3 and OP5 all feeding into it directly. For now leave OP5 at zero and use the other two.

Try this

Set OP2 to level 65 with FreqCoarse 1, and OP3 to level 50 with FreqCoarse 7.

Take each one down to zero in turn and listen to what disappears. OP2 gives you the body of the sound; OP3 adds a thin, reedy edge on top. Together they make something neither could produce alone.

Now give them different envelopes, so the edge fades before the body does.

Two modulators side by side on one carrier behave pretty much as you'd hope: each contributes its own character and you hear both. That makes them predictable and easy to work with, which is why this arrangement turns up in a great many factory sounds.

There is another way to connect three operators, and it's much less well behaved. That's the next chapter.

What you have at the end of this chapter

An electric piano, a bell, and a working understanding of the only idea in FM synthesis that really matters: some operators are heard, some operators change the ones that are heard, and the difference between them is nothing but where they're plugged in.

Save two or three of your favourites before you move on.

Chapter 5 - Algorithms

The wiring diagram, and what you can guess from its shape.

You've been using algorithms since chapter one, when you sat there turning the ALGORITHM knob and listening to a preset fall apart and put itself back together. Now for what was actually happening.

5.1 Reading the picture

There are thirty-two of them, and turning the ALGORITHM knob shows you a diagram of the one you've landed on. Every diagram is made of the same parts:

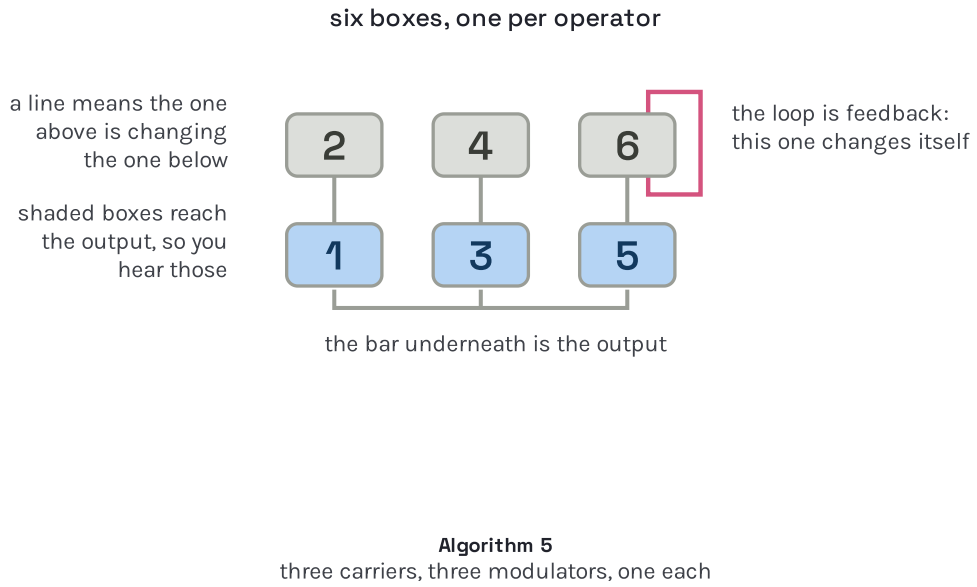


Figure 5.1 - How to read an algorithm.

The whole chart is at the end of this chapter. It is meaningless until you know what the boxes mean, and alarming if you meet it cold, so leave it until you've read the rest of the chapter.

5.2 Tall and thin, short and fat

Look down the chart and you'll notice the shapes fall into two rough families, and you can predict a great deal from which family an algorithm belongs to.

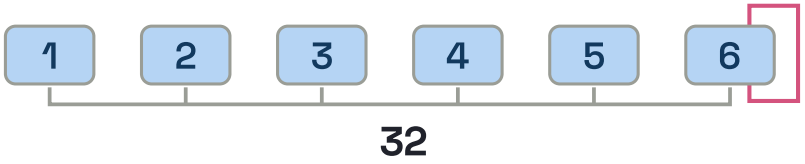


Figure 5.2 - Algorithm 32. Six carriers, no modulation at all.

Short and fat means lots of carriers side by side. Algorithm 32 is the extreme case: all six operators are carriers, none of them modulating anything. Six sine waves, added together, and nothing else. That's exactly how a pipe organ works - separate pipes at different pitches, sounding at once - and it's what these algorithms are good for. Smooth, clean, blendy sounds. Organs, strings, choirs, pads.

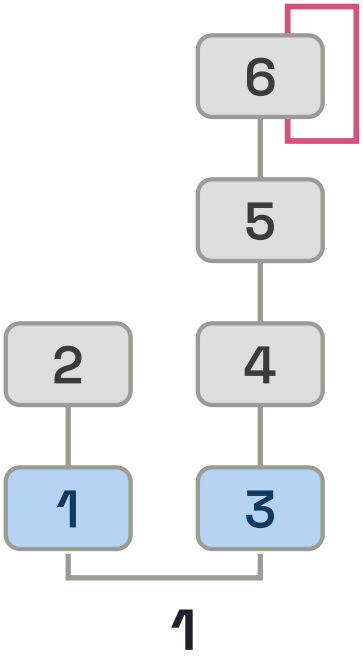


Figure 5.3 - Algorithm 1.

Tall and thin means operators stacked in a chain, each one modulating the next. Algorithm 1 has a stack four operators deep. Each stage multiplies the complexity of the one below it, so the results are dense, bright, often metallic. Bells, chimes, aggressive basses, sound effects.

Most of the thirty-two sit somewhere in between, and that's usually where you want to be - a couple of carriers for body, a stack for character.

> **Try this**
>
> Take one of the presets you liked in chapter one and sweep the ALGORITHM knob again, exactly as you did then. But this time watch the diagram as it goes past.
>
> You should now be able to see, before you hear it, roughly what's about to happen. That's the whole point of the chart.

5.3 Stacking modulators

In the last chapter you put two modulators side by side. Now for the other way of connecting three operators.

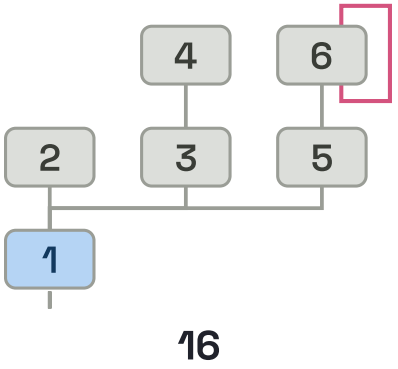


Figure 5.4 - Algorithm 16 again.

Stay on algorithm 16 and look at where OP4 goes: it doesn't feed OP1 at all. It feeds OP3, which feeds OP1.

> **Try this**
>
> Set OP3 to level 60 with FreqCoarse 1, and leave OP2 and OP5 at zero. Familiar territory - one modulator, one carrier.
>
> Now bring OP4's level up slowly from zero.
>
> OP4 is modulating a modulator. You aren't making the sound brighter, you're making the *brightening* more complicated, and the result gets strange much faster than it did in chapter four. Small movements make big differences.

Modulators side by side add up in ways you can predict. Modulators stacked on top of each other multiply in ways that are extremely hard to predict, even for people who have been doing this for decades.

Which is not a reason to avoid them. It's a reason to turn the knob slowly and keep the ones you like.

5.4 Feedback

One operator in every algorithm has a loop drawn round it, and there's a Feedback control on the patch Globe page running from 0 to 7.

EDIT GLO page 1/2 - Algorithm, Feedback, Osc Sync and Transpose, on knobs 1 to 4.

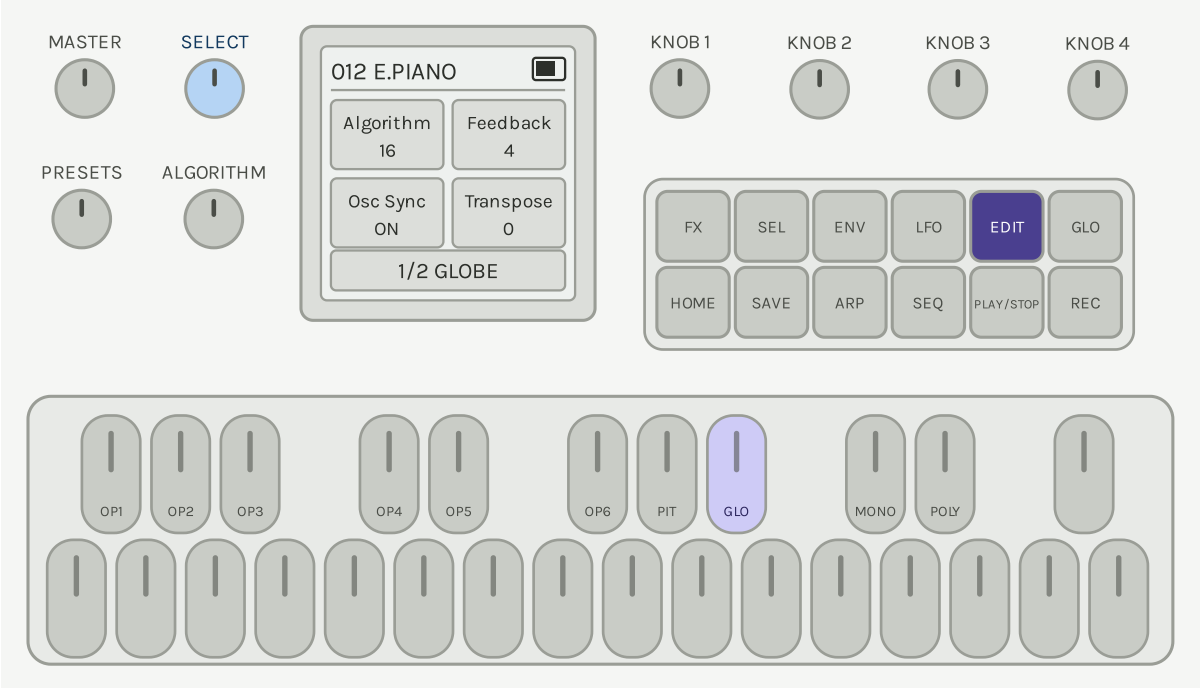


Figure 5.5 - The patch Globe page, reached through EDIT.

> **Algorithm** (KNOB 1, 1-32) - which of the thirty-two wirings the sound uses.

> **Feedback** (KNOB 2, 0-7) - how hard the looped operator modulates itself.

> **Osc Sync** (KNOB 3, On or Off) - whether the operators restart their waveforms on each note. Chapter seven has this one.

> **Transpose** (KNOB 4) - shifts the whole instrument in semitones.

Feedback means an operator modulating itself. Turn it up gradually and the operator's pure sine wave gets progressively more jagged - first brighter, then buzzy, and at the top of the range it breaks down into noise.

It's a small control with a big range of effect, and it's how you get roaring basses, hissing cymbals and wind noises out of an instrument that only contains sine waves. As feedback rises the FM-1 pulls the overall output level down a little to stop things getting out of hand.

> **Try this**
> Set algorithm 16. The loop is round OP6, OP6 modulates OP5, OP5 modulates OP1, and OP1 is the only carrier.
>
> So three operators have to be up before the loop is audible at all: OP1, because it is the one that reaches your ears; OP5, because it is what OP6 is modulating; and OP6 itself. Turn OP2, OP3 and OP4 down out of the way.
>
> Now set feedback to 0 and step it up one number at a time. What is changing is OP6's own waveform, and you hear it as a change in the brightness of OP1.
>
> Seven settings, and you can hear all seven. There is no other control on this instrument with that few positions and that much difference between them.

5.5 An honest word about all this

Here's something the manuals tend not to say.

People who are extremely good at FM synthesis - people who have made a living at it for thirty years - still spend a lot of their time turning knobs and going "ooh, what's that". The theory tells you roughly where to look. It does not tell you what you'll find. Nobody sits down and calculates their way to a good bell sound.

What the theory buys you is this: when something sounds nearly right, you know which knob to reach for. And when something sounds wonderful by accident, you can work out why, which means you can do it again on purpose. That's genuinely all of it, and it's enough.

5.6 All thirty-two

Here they are. You are not meant to learn them, and nobody has. What you are meant to be able to do is glance at one and guess roughly what it will sound like, which after this chapter you can.

Carriers are shaded. The pink line is the feedback loop, and it is worth looking for: twenty of the thirty-two come in pairs or threes with identical wiring, differing only in where that loop sits.

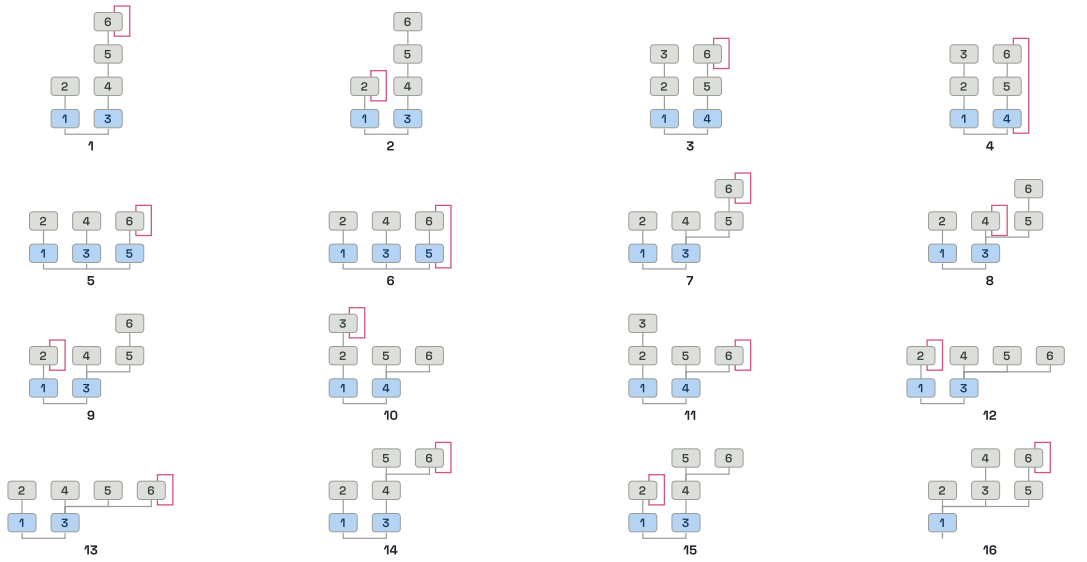


Figure 5.6 - Algorithms 1 to 16.

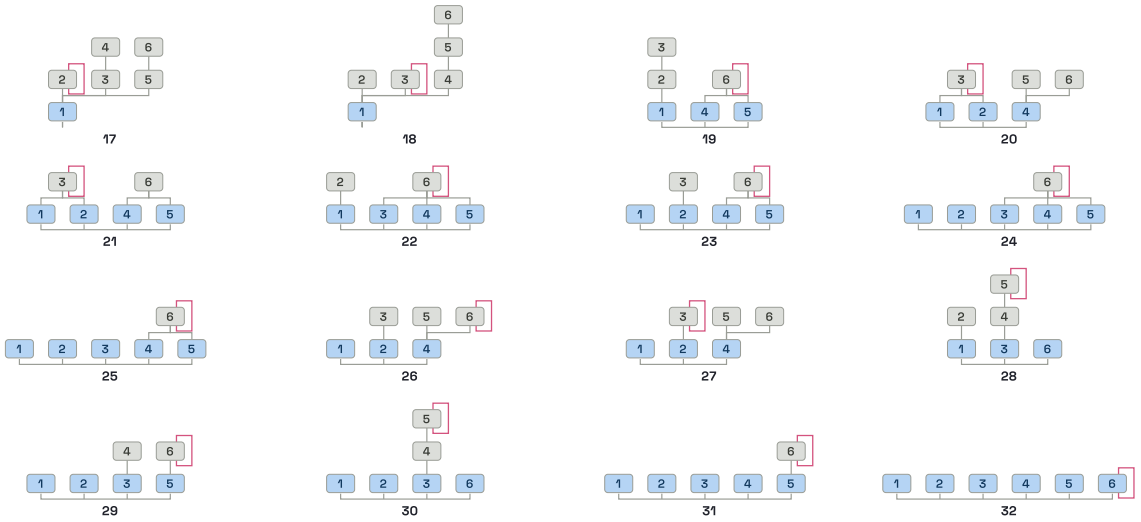


Figure 5.7 - Algorithms 17 to 32.

Where you've got to

You can start a sound from nothing, shape how it grows and fades, make it bright or dull or metallic or alive, change how that brightness moves over time, and read the wiring diagram well enough to guess what an unfamiliar algorithm will do.

That's the synthesis engine. Everything after this - the effects, the arpeggiator, the sequencer - is about what you do with the sounds once you've made them.

Chapter 6 - Effects

The sound leaves the synthesizer and then six things happen to it.

Everything so far has been about making a sound. This chapter isn't. This chapter is about what happens to that sound on its way out, and it's the cheapest improvement in the whole instrument - a dull patch with the right reverb on it will beat a clever patch with none.

It also works on absolutely anything. You don't need to understand operators or envelopes for this chapter to pay off. If you skipped straight here from chapter one, that's fine.

6.1 The chain

Press FX. The screen shows six boxes in two rows of three, and each one is an effect: Filter, Reverb, Delay, Distortion, Chorus, Phaser.

The word to hang on to is *chain*. These six aren't a menu you pick from - they're a queue. Your sound goes into the first box, whatever comes out of that goes into the second, and so on through all six before it reaches your ears. Every box is either doing something or standing aside.

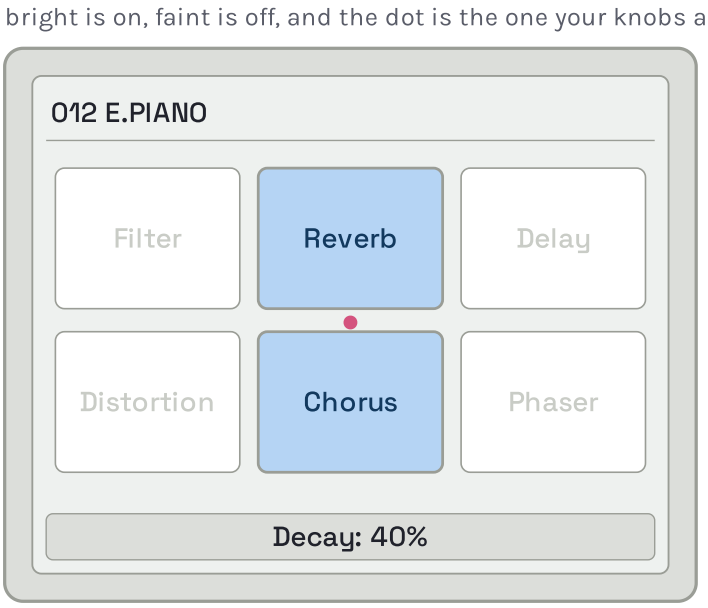


Figure 6.1 - The chain, on screen. Bright is on, faint is off.

Two things tell you the state of each box. A bright box is switched on; a dim box is switched off. And a dot blinks underneath whichever box you've currently got selected - that's the one your knobs are pointed at.

1. Press FX to get in.
2. Turn SELECT to move the blinking dot from box to box.
3. Turn KNOB 4 clockwise to switch that effect on, anticlockwise to switch it off.
4. Turn KNOB 1, 2 and 3 to change how it sounds. The bottom bar tells you what each knob is doing and what value it's on.
5. Press FX again to leave.

KNOB 4 is always the on/off switch, in every one of the six - which is worth knowing, because it means you can go along the whole chain switching things in and out without ever looking up what the knobs do.

> **Try this first**
>
> Load a preset you like. Press FX, select Reverb, and turn KNOB 4 clockwise to switch it on. Play a chord and hold it.
>
> Now turn KNOB 4 back the other way, and play the same chord again.
>
> That's the difference between an instrument that sounds like it's in a room and one that sounds like it's in your head.

6.2 What the six actually do

KNOB 4 is the on/off switch on every one of the six - clockwise on, anticlockwise off - so it isn't listed again below. Two of them, Reverb and Filter, have a type to choose as well, and on those KNOB 1 picks it.

Descriptions first, numbers afterwards. Two of the six - Filter and Reverb - have types to choose between, and on those two KNOB 1 picks the type. The other four go straight to parameters.

6.2.1 Reverb

Puts the sound in a space. Start here.

> **Type** (KNOB 1, Room, Hall or Plate) - which space. Room is small and believable. Hall is big and obvious. Plate is the one that sounds like an old record: a metal sheet in a box, historically, and it flatters bells and electric pianos in a way the other two don't.

> **Decay** (KNOB 2, 0-100%) - how long it takes to die away.

> **Mix** (KNOB 3, 0-100%) - how much of it you hear.

6.2.2 Filter

Removes part of the sound. Quietly the most powerful one here.

> **Type** (KNOB 1, low, band or high pass) - which part it removes. Low pass keeps the low end and throws away the top, which makes things sound warm, muffled, distant, underwater. High pass does the opposite and makes them thin and tinny. Band pass keeps a slice in the middle and discards both ends, which is how you get telephones and old radios.

> **CutOff** (KNOB 2, 100 Hz to 20,000 Hz) - where it starts cutting.

> **Q** (KNOB 3, 0-100 in tens) - the resonance: how much the filter shows off at the point where it's cutting. Low Q and you barely notice it working. High Q and it starts to whistle at the cutoff frequency. This is the one effect parameter that still moves in ten steps rather than a hundred, so you cannot set it to 45. It shows 0 to 100 but there are only eleven positions.

6.2.3 Delay

Repeats the sound. Echo.

> **Rate** (KNOB 1, 0-100%) - how fast the repeats come.

> **Decay** (KNOB 2, 0-100%) - how many you get before they fade out.

> **Mix** (KNOB 3, 0-100%) - how much of it you hear.

Short rate and low mix gives you a thickening that doesn't announce itself. Long rate and high decay gives you the thing that repeats after you've stopped playing, which is enormous fun and gets tiring quite quickly.

6.2.4 Distortion

Damages the sound on purpose.

> **Gain** (KNOB 1, 0-100%) - how hard you're driving it.
>
> **Tone** (KNOB 2, 0-100%) - bright or dark.
>
> **Level** (KNOB 3, 0-100%) - your output volume, and you will need it.

Distortion makes things louder as well as dirtier, which is why Level is there. Turn the gain up and pull the level down to compensate. On an FM sound it does something more interesting than it does on a guitar, because there's more going on in the sound for it to mangle.

6.2.5 Chorus

Makes one instrument sound like several slightly out-of-tune ones.

> **Depth** (KNOB 1, 0-100%) - how far out of tune the copies drift.
>
> **Freq** (KNOB 2, 0-100%) - how quickly they wander.
>
> **Mix** (KNOB 3, 0-100%) - how much of it you hear.

It's the sound of every electric piano recorded in the nineteen eighties, and the FM-1 is exactly the instrument that wants it.

6.2.6 Phaser

The whooshing one.

> **Depth** (KNOB 1, 0-100%) - how strong the sweep is.
>
> **Freq** (KNOB 2, 0-100%) - how fast it sweeps.
>
> **Mix** (KNOB 3, 0-100%) - how much of it you hear.

Same controls as chorus, quite a different result: a sweeping, hollow, rotating sort of movement. Sparingly on pads, it's lovely. On everything at once, it's nineteen seventy-three.

6.3 Order matters

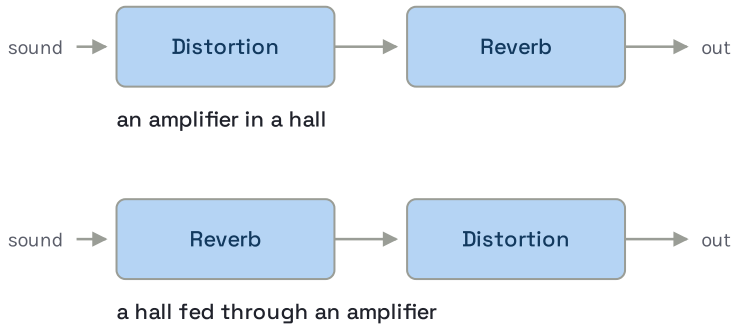


Figure 6.2 - The same two effects, both ways round.

Six effects in a queue, and you can change the order of the queue. This is less fussy than it sounds and the reason is easy to hear: distortion after reverb damages the room, and reverb after distortion puts the damaged sound in a room. Those are not the same thing.

1. In FX mode, turn SELECT until the dot is under the effect you want to move.
2. Press SEL. The dot stops blinking and goes solid, and the SEL light comes on - you've grabbed it.
3. Turn SELECT. The grabbed effect swaps places with its neighbour, and keeps swapping as you turn.
4. Press SEL again to drop it. The light goes out.

SEL only does this inside FX mode, which is a small mercy - one fewer button doing two jobs.

> **Hear the difference**
>
> Switch on Distortion and Reverb, and nothing else. Put Distortion first in the chain and play a few notes. Then grab Reverb and move it in front of Distortion, and play the same notes.
>
> One of those sounds like an amplifier in a hall. The other sounds like a hall being fed through an amplifier.
>
> Wind both effects up hard before you try it - Distortion's Gain high, Reverb's Mix high. At polite settings the difference is there but slight, slight enough that you will suspect yourself of imagining it. Pushed, it is obvious. Neither order is wrong, and once you can hear it at the extremes you will start hearing it at sensible settings too.

6.4 A place to start

If you want somewhere to begin rather than six unknowns, this is a decent general-purpose setting for almost any sound.

- Reverb on, Plate, Decay around 25, Mix around 10 to 15.
- Chorus on, Depth around 30, Freq around 20, Mix around 30.
- Everything else off.

It won't sound like anything in particular, which is the intention - it should sound like the patch you already had, but as though someone recorded it properly.

> **Switch them off before you judge a patch**
>
> Effects are flattering. That's their job. It also means they'll hide problems in a sound - a patch with too long an attack, or an envelope that doesn't quite release, will sound fine drenched in reverb and wrong the moment you take it away.
>
> When you're building a sound from chapter three onwards, build it dry. Add effects at the end, once you're happy with the thing underneath.

Nothing in this chapter changes the synthesizer - every operator, envelope and algorithm is exactly where you left it. Effects sit on top of the sound, and you can rip them all off again without having lost any work.

Everything in FX belongs to the patch

Which effects are on, what their parameters are set to, and the order they run in are all stored with the patch, exactly like the operator settings in chapter two. Change preset without saving and you lose them. Press SAVE and they come back with the sound.

So effects here are part of sound design rather than a performance layer sitting over the top, which is why this chapter is where it is.

An effect that is switched off is skipped rather than passed through. Move a switched-off effect around the chain and nothing changes at all, because position only matters between effects that are on.

Chapter 7 - Movement

An oscillator too slow to hear doesn’t make a sound, it makes other things wobble.

Every sound you've built so far does the same thing on every note. Press a key, the envelopes run, the note comes out, and if you hold it down it sits there being exactly what it is. Real instruments don't. A held violin note shivers. A singer's long note starts straight and then starts to move. A Rhodes played hard doesn't just get louder, it gets angrier.

This chapter is about the two things on the FM-1 that put movement into a sound after the note has started: the LFO, and the ENV button. It ends with a third kind of movement, glide, which happens between notes rather than during them.

7.1 The LFO

LFO stands for low frequency oscillator, and it's the most literal name in synthesis. It's an oscillator, exactly like the six you've been working with, except it runs so slowly you can't hear it as a pitch. A few cycles a second rather than a few hundred.

Because you can't hear it, it isn't used to make sound. It's used to move something else. Point it at pitch and you get vibrato. Point it at volume and you get tremolo. That's the whole idea, and everything else in this section is a detail of it.

LFO - press it to get in, press it again to go back to Home. There are two pages: page 1 is the sound of it, page 2 is the behaviour of it.

7.1.1 Page 1

LFO page 1/2 - Wave, Speed, PMD and AMD, on knobs 1 to 4.

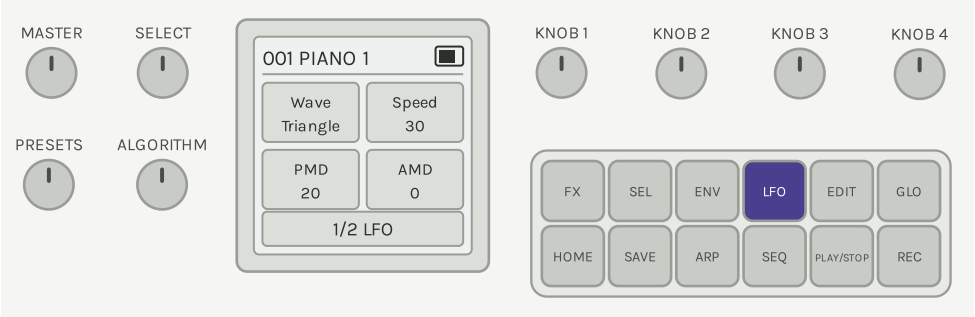


Figure 7.1 - LFO, page 1.

PMD (KNOB 3, 0–99) - Pitch Modulation Depth: how far the LFO pushes the pitch up and down. Small amounts are vibrato. Large amounts are a siren.

AMD (KNOB 4, 0–99) - Amplitude Modulation Depth: how far it pushes the volume up and down. Small amounts are a slight throb. Large amounts are a helicopter.

Speed (KNOB 2, 0–99) - how fast the wobble is. Human vibrato sits around five or six cycles a second, which puts it somewhere around 35 to 40 on this dial. The scale is exponential, doubling roughly every twenty steps: about 2 Hz at 10, 4 Hz at 30 and 8 Hz at 50, and past that too fast to count. Equal turns of the knob buy equal musical intervals of speed, which is why it feels even across its range.

Wave (KNOB 1) - the shape of the wobble. There are six.

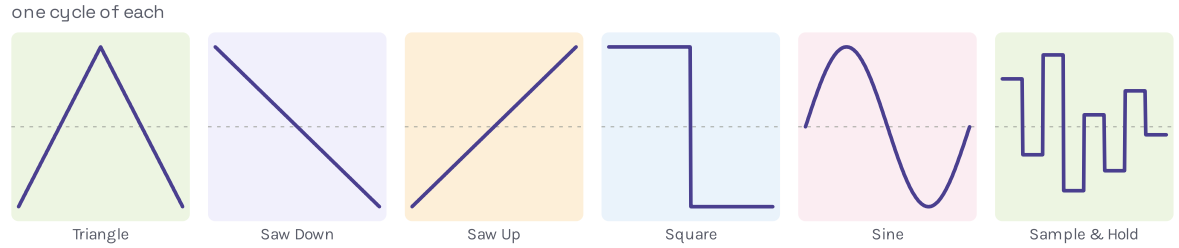


Figure 7.2 - One cycle of each of the six shapes.

The first two both go smoothly up and down, and for vibrato you will not be able to tell them apart - the sine is fractionally rounder at the turns. Either is right.

The square jumps: straight to the top, straight to the bottom, nothing in between. On pitch that gives you a trill between two notes rather than a wobble. On volume, a hard on-off stutter.

The two sawtooths climb and drop back, or fall and jump back. On pitch, the falling one is the sound of something powering off, over and over.

The last one picks a random value, sits on it, then picks another. It's the sound of nineteen-seventies computers thinking, it has almost no use in an imitative sound, and it is enormous fun everywhere else.

> **Try this first**
>
> Load any preset. Press LFO. Turn KNOB 3 (PMD) up to about 20, and KNOB 2 (Speed) to about 30. Play a note and hold it.
>
> Now wind PMD slowly down towards zero while the note is still sounding, and find the point where you stop noticing the wobble and start just hearing a slightly livelier instrument. That point is where you want to be most of the time.

At the very top of its range the LFO arrives at the edge of audio rate. Wind Speed to 99 with PMD high and the vibrato stops sounding like movement and starts sounding like a tone colour. That is not a coincidence or a fault. It is the same thing the operators do to each other in chapters four and five, happening slowly enough that you can hear it cross over.

7.1.2 Page 2, and the parameter that catches people out

LFO page 2/2 - Delay, Pitch Sens, Lfo Sync and Osc Sync, on knobs 1 to 4.

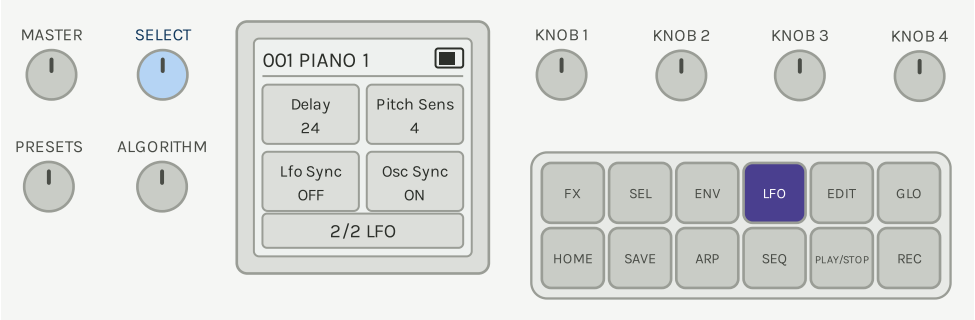


Figure 7.3 - LFO, page 2.

Delay (KNOB 1, 0–99) - how long the LFO waits after you press the key before it starts. This is the single setting that makes vibrato sound like a person rather than a machine. Nobody plays a note with the vibrato already running - a violinist arrives on the note, and then it starts to move. Put a delay on it and the same wobble that sounded cheap sounds expressive.

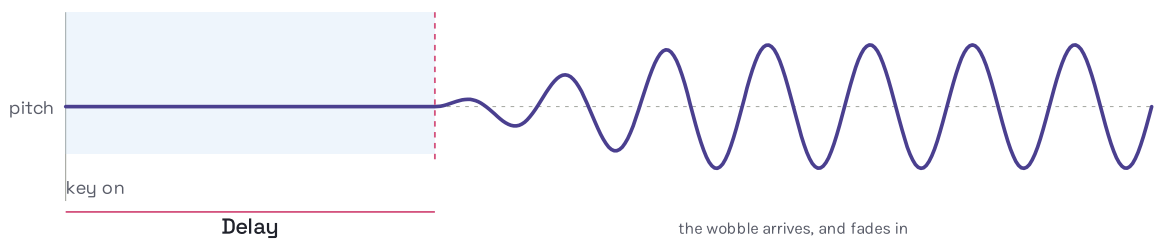


Figure 7.4 - The delay period, then the wobble fading in.

Pitch Sens (KNOB 2, 0–7) - the one that catches people out. On the DX7, from which all of this descends, the LFO's pitch modulation has two controls in series: the depth on the LFO itself, and a per-voice sensitivity deciding how much of that depth gets through. At zero sensitivity you can put PMD at 99 and hear absolutely nothing. The FM-1 works the same way. With Pitch Sens at 0 there is no vibrato at all, however high you set PMD, and raising Pitch Sens from 1 brings it in. If you have set PMD and nothing is happening, this is why. AMD has a gate too, arranged differently. Pitch modulation is gated once, globally, by Pitch Sens here. Amplitude modulation is gated per operator, by A ModSens on each operator's page 4, and that runs 0 to 3 rather than 0 to 99. So vibrato is switched on in one place, and tremolo has to be enabled on every operator you want it to affect.

Lfo Sync (KNOB 3) - whether the LFO restarts from the beginning of its cycle on each note, or runs continuously in the background. Off, every note catches the wobble at a different point, which sounds natural on single notes and slightly chaotic on chords. On, every note starts its wobble together, which sounds deliberate on chords and slightly mechanical on melodies.

Osc Sync (KNOB 4) - the same question for the operators themselves: whether they restart their waveforms on each note or carry on from wherever they happened to be. The audible difference is in the very first instant of the note, and it's subtle. This same setting also appears on the patch Globe page, EDIT then GLO, page 1. It is one setting shown in two places, so changing it here changes it there.

7.2 The ENV button

Here is a thing the manual mentions in passing and does not explain.

There is a second envelope, separate from all six operator envelopes, and it is a plain four-stage ADSR. It is switched off by default. Press ENV and it lights, and you are editing it. Press ENV again and the whole function switches off.

ENV - Attack, Decay, Sustain and Release, on knobs 1 to 4.

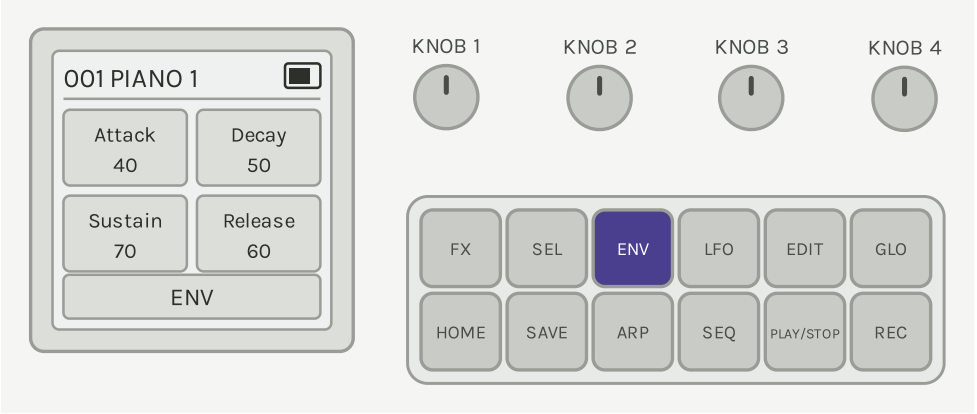


Figure 7.5 - ENV.

All four run 0 to 100%, and each shows as a bar graph rather than a number.

> **Attack** (KNOB 1) - how long the sound takes to reach full volume. 0% instant, 100% a very slow fade in.

> **Decay** (KNOB 2) - how quickly it falls from that peak to the sustain level.

> **Sustain** (KNOB 3) - the level it holds at while you keep the key down.

> **Release** (KNOB 4) - how long it takes to fade out after you let go. 0% is an immediate cut, 100% a long tail.

If chapter three has done its job, none of that is new to you. It is the same idea as an operator envelope with two of the stages taken away, and it is the shape that most synthesizers in the world use.

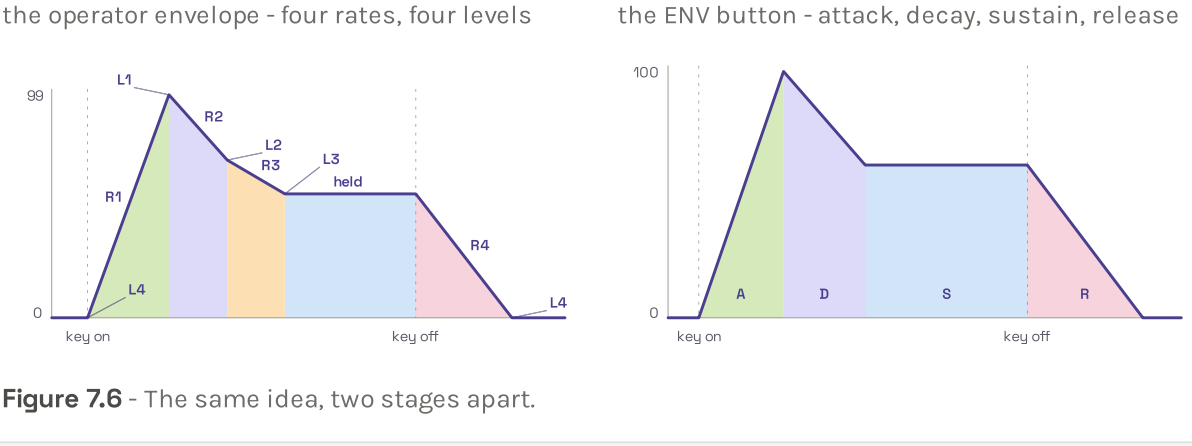


Figure 7.6 - The same idea, two stages apart.

Two envelopes, and how they get on

You now have, potentially, seven envelopes running at once: one per operator, plus this one. This one applies to the whole output after the operators have had their say, and the two multiply together.

That has a consequence worth holding on to, and it comes with a trap. The two envelopes count in opposite directions. On ENV, a higher Attack means a slower attack, because it is a duration. On an operator, a higher Rate 1 means a faster attack, because it is a speed. So winding both up moves the sound in opposite directions.

With that straight, the rule is simple enough. ENV can make an attack slower than the operators make it, and it can fade out a note the operators are still holding. It cannot do the reverse. Set ENV Attack faster than the operator envelope and nothing happens at all, because the operator's own fade is already the later of the two and it is what you hear.

The same goes for Release. If the operators have already dropped the sound to silence when you lift your hands, there is nothing left for ENV to fade, and winding Release up does nothing.

There is a second reason to go carefully with it. The operator envelopes are four rates and four levels, a shape defined by how fast it moves between points. ENV is attack, decay, sustain and release, a shape defined by durations and a held level. Those are two different ways of describing the same thing, and using both at once means keeping both in your head and imagining their product, with nothing on screen to show you the result.

So build sounds with the operator envelopes, the way chapters three to five describe. Reach for ENV when you cannot get what you want any other way, which usually means modifying someone else's preset without learning how it is built.

7.3 What ENV is genuinely good for

Two things, both of them practical rather than creative.

The first is rescuing a preset. You've found a sound you like from chapter one, and the only thing wrong with it is that it stops too abruptly when you lift your hands. Press ENV, wind Release up, done. You have not had to learn what the preset is made of.

That works only if the preset is still holding some level when you lift your hands. If its operators drop straight to silence, ENV has nothing left to fade and Release does nothing at all. A fair number of the presets that stop too abruptly are abrupt for exactly that reason, so if it doesn't work, that is why.

The second is turning anything into a pad. Attack up, Sustain high, Release long, and a plucked sound becomes something that swells and hangs. It's a crude effect and it works on absolutely anything.

Make a preset breathe

Find a bright, short sound in the presets - something plucky. Press ENV. Wind Attack up to around 40, Sustain to around 70, and Release to around 60.

Play a chord and let go of it. The same sound is now doing something completely different, and you have changed nothing about how it is made.

Press ENV once more to switch the whole thing off, and the plucky sound comes back exactly as it was.

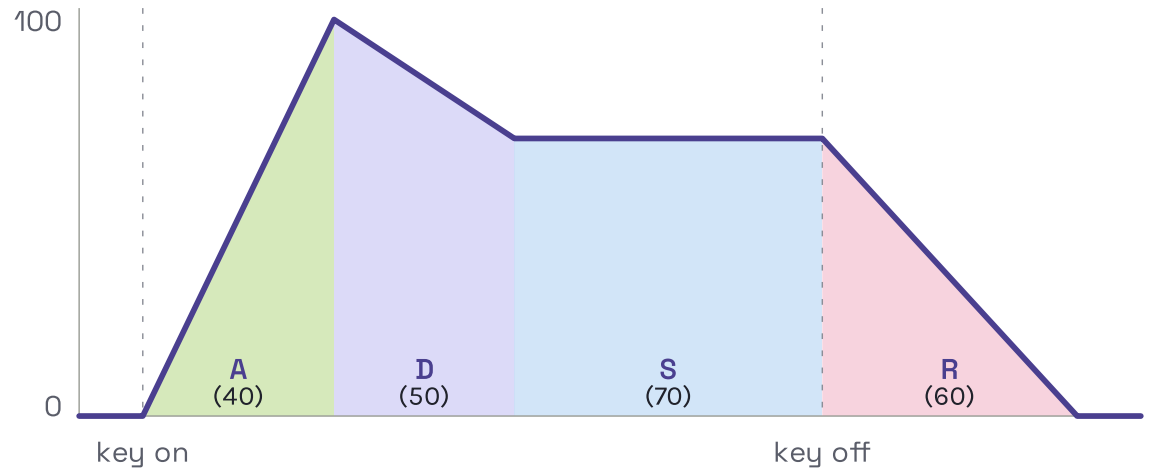


Figure 7.7 - The settings from the box above.

LFO settings save with the patch, like the effects in chapter six and everything else you have edited so far. ENV does too.

7.4 Glide

Glide is the third kind of movement here, and the only one that happens between notes rather than inside them. Play one note, then another, and the pitch slides across instead of jumping.

It sits on the instrument Globe page, which is the GLO button pressed on its own rather than EDIT and then GLO.

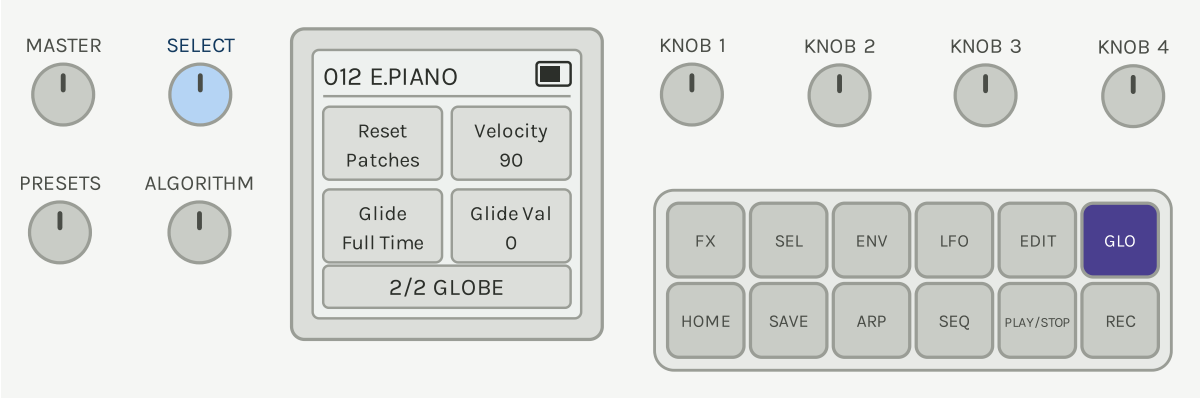


Figure 7.8 - The instrument Globe page, from the GLO button alone.

GLO page 2/2 - the two glide controls are on knobs 3 and 4, past Reset Patches and Velocity. `Glide` on KNOB 3 chooses whether the slide happens on every note or only when you hold one note down and play another. `Glide Val` on KNOB 4 sets how long it takes, from 0 to 1 second.

Both are greyed out until you press MONO. Glide works on one note at a time, and MONO and POLY are two of the labelled keys in edit mode. Press MONO and both come alive.

MONO saves with the patch, so a glide sound can be stored as a preset and comes back as one.

Please do not read this text

Chapter 8 - Playing across the keyboard

A real instrument doesn’t behave the same way at both ends of itself.

Play the bottom note of a piano and the top note of the same piano. They are not the same sound an octave or six apart. The bottom note is long, dark and complicated. The top note is short, bright and thin. Nobody designed that on purpose - it falls out of the fact that one string is long and thick and the other is short and tight.

An FM patch has no strings, so nothing falls out of anything. By default, the sound you built in chapter four does exactly the same thing at both ends of the keyboard, and that uniformity is one of the main reasons a patch can be technically correct and still sound fake.

This chapter is about the settings that break the uniformity. It is the fiddliest part of the instrument, and it gives the least reward per minute spent, which is why it's this far back in the book. It is also the difference between a patch that's fine and a patch that's convincing.

Everything here lives on the later pages of the operator editor, and it is all *per operator*. That is the important part. You are not adjusting the sound; you are adjusting one operator's contribution to the sound, and the interesting effects come from doing it to modulators rather than carriers.

8.1 Sensitivity and mode

EDIT OP1 page 4/6 - A ModSens, KeyVelocity and OscMode, on knobs 1 to 3.

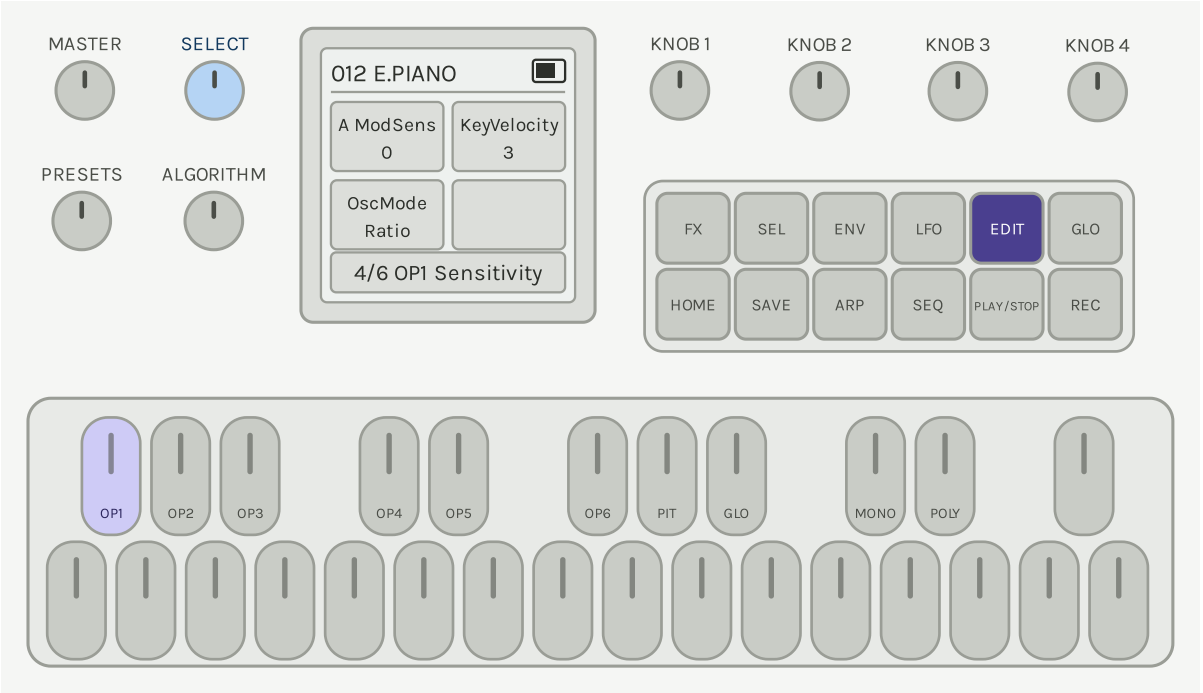


Figure 8.1 - Operator page 4.

> **KeyVelocity** (KNOB 2, 0–7) - how much the force of your playing affects this operator's output level. At 0, playing hard and playing softly produce identical results from this operator. At 7, the difference is dramatic.

> **A ModSens** (KNOB 1, 0–3) - how much of the LFO's amplitude modulation reaches this particular operator. At 0, the tremolo you set up in chapter seven passes this operator by completely.

> **OscMode** (KNOB 3, Ratio or Fixed) - Ratio is normal: the operator's frequency is a multiple of the note you played, so the whole sound moves up and down the keyboard together. Fixed means the operator ignores the note entirely and sits at one frequency wherever you play.

Two of those three do something more interesting than they sound, and in both cases the trick is the same: put them on a modulator rather than a carrier.

Velocity on a carrier is a volume control - play harder, that note is louder. Velocity on a modulator is a brightness control: play harder, that operator modulates more strongly, and the sound gets brighter as well as louder. That second one is what real instruments do. Hit

a piano key harder and it doesn't just get louder, it gets harsher. Blow a trumpet harder and it gets brassier. Set velocity on your modulators, leave your carriers alone, and you get an instrument that changes character with how hard you play it.

Amplitude modulation works the same way round. On a carrier you get a wobble in volume, which is plain tremolo. On a modulator you get a wobble in brightness - the sound shimmers rather than pulses, which is a much more musical effect, and it's the reason this setting is per operator instead of global.

OscMode is the odd one out on this page: it belongs with chapter four and has been sitting here all along. Set a modulator to Fixed and give it a very short envelope, and you get a click or a thump that is identical on every note - the sound of a hammer, or a key mechanism, or a breath. Real instruments have exactly this. The noise a piano makes is not transposed when you play higher, because it comes from the mechanism rather than the string. It's a small detail and it's very convincing.

> ### The keyboard does not sense velocity
>
> The FM-1's own 27 keys all play at the same strength, however hard you hit them. What sets that strength is the `Velocity` setting on the GLO button's page 2, KNOB 2, running 1 to 127. Set it to 30 and the whole instrument plays as though you were touching it gently; set it to 120 and everything sounds struck hard.
>
> So this section is not wasted, it just cannot be demonstrated with your fingers. Wherever it asks you to play softly and then firmly, set Global Velocity low and play, then set it high and play again. That isolates the variable rather better than hands would anyway.
>
> Play the FM-1 from another keyboard over MIDI and velocity works normally, note by note, and Global Velocity has no effect on it whatsoever. The two are independent.

8.2 Keyboard scaling

Two pages of it, and it does one thing: it makes an operator get quieter, or louder, the further you play from a chosen point on the keyboard.

> **If your unit disagrees, check the firmware**
>
> The page numbers in this chapter were read off a real unit on firmware v15 and are right for it. They have not always been: a version of the firmware in circulation ordered the six operator pages differently. If your unit puts these controls somewhere else, find out which firmware you are on before assuming the book is wrong.

8.2.1 Page 5 - where and how much

EDIT OP1 page 5/6 - L Depth, R Depth and BreakPoint, on knobs 1 to 3.

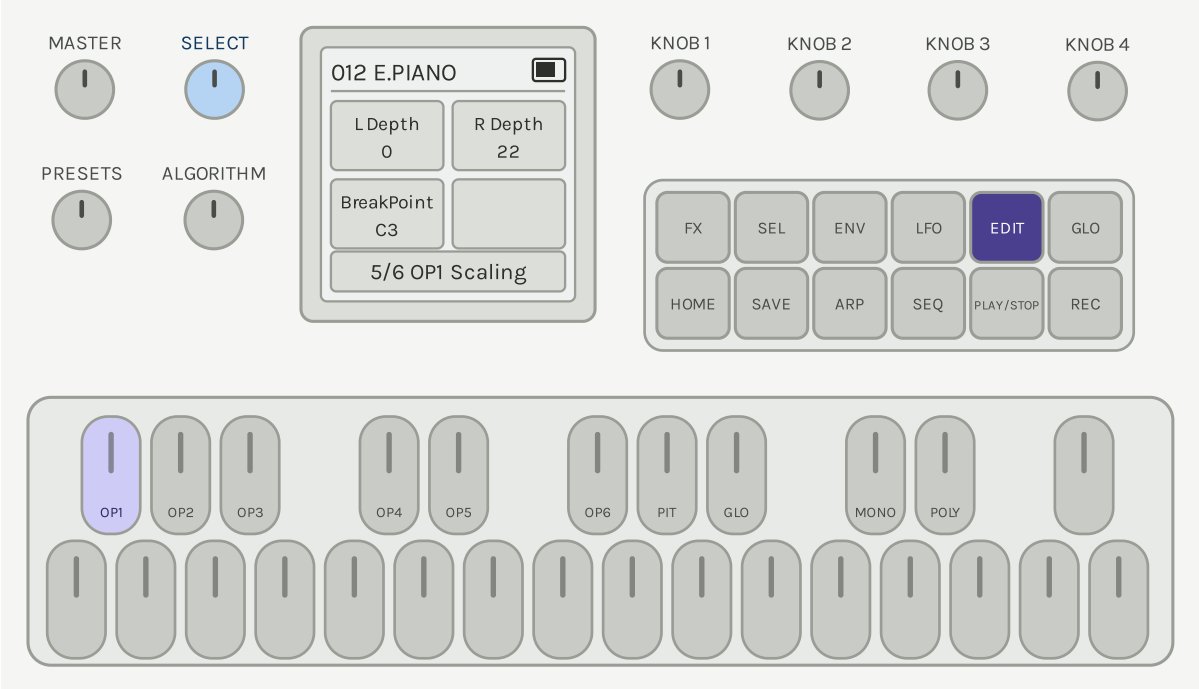


Figure 8.2 - Operator page 5.

BreakPoint (KNOB 3, A-1 to C7) - a note on the keyboard, which you choose. Everything below it is the left side; everything above it is the right side. It is set by turning KNOB 3 and only by turning KNOB 3: playing a key while this page is showing does nothing. If you have come from a DX7, where you hold a button and play the note you want, that is worth knowing before you decide the control is broken.

L Depth (KNOB 1, 0–99) - how strong the effect is on the left side. At 0 that side does nothing at all, whatever curve you pick.

R Depth (KNOB 2, 0–99) - the same, for the right side.

8.2.2 Page 6 - what shape, and how fast

EDIT OP1 page 6/6 - L Curve, R Curve and RateScale, on knobs 1 to 3.

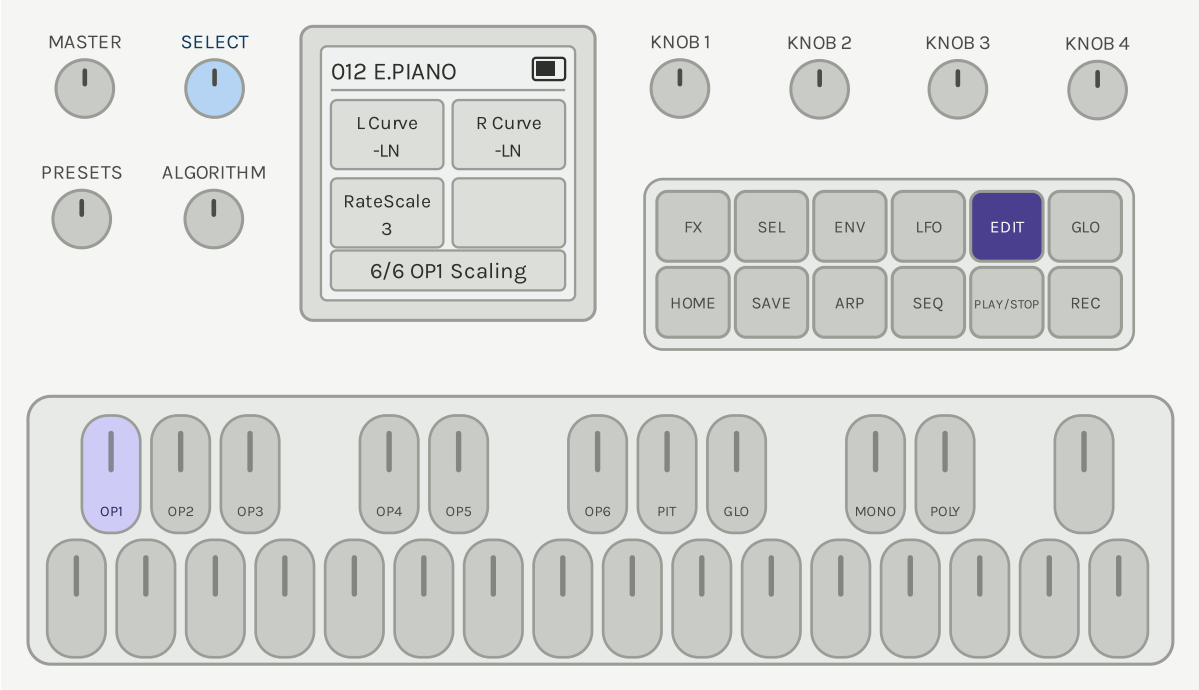


Figure 8.3 - Operator page 6.

L Curve (KNOB 1) - the shape the left side takes: -LN, -EX, +LN or +EX.

R Curve (KNOB 2) - the same four choices, for the right side.

RateScale (KNOB 3, 0–7) - how much faster this operator's envelope runs as you play higher up the keyboard. Section 8.3 is about this one.

The minus curves make the operator quieter as you move away from the break point; the plus curves make it louder. LN is linear, meaning it changes evenly; EX is exponential, meaning it changes slowly at first and then rapidly.

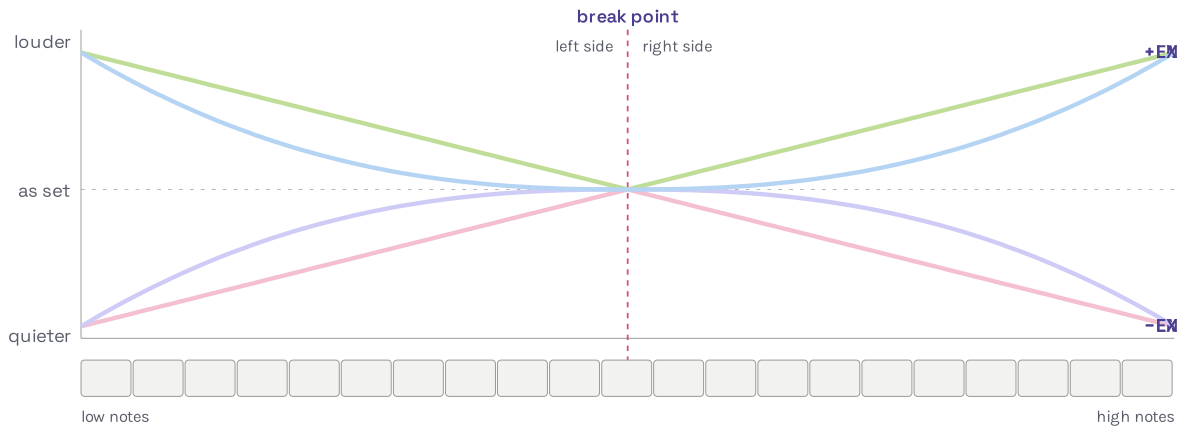


Figure 8.4 - Break point, depth and curve. The four curve types, about one break point.

The classic use is a modulator with a break point somewhere in the middle of the keyboard, a minus curve on the right, and a moderate right depth. Now the top of the keyboard is less brightly modulated than the bottom, which is what nearly every acoustic instrument does - high notes are purer, low notes are richer.

8.3 Rate scaling, which is the one to set first

Everything above changes how loud an operator is at different pitches. This changes how *fast* its envelope runs at different pitches.

Turn it up and notes at the top of the keyboard decay faster than notes at the bottom, which is the piano fact from the first paragraph of this chapter, made into a single knob. It is the highest return on effort of anything in this chapter, and if you are going to set one thing here and ignore the rest, set this.

It reaches the operator envelopes and nothing else. The pitch envelope runs at the same speed wherever you play, which follows from RateScale being a per-operator setting while the pitch envelope is not.

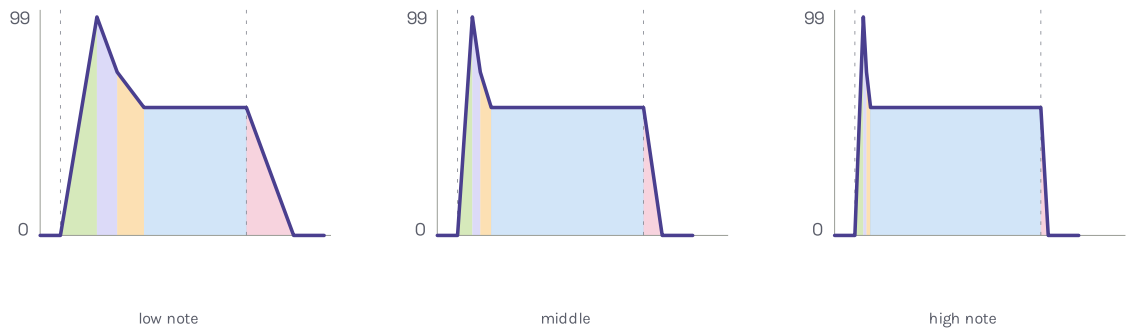


Figure 8.5 - One envelope at three pitches. Higher notes decay faster.

Hear rate scaling on its own

Take your electric piano from chapter four. Play a low note and let it ring, then play a note two octaves up and let that ring. Time them roughly. The high note outlasts the low one, which is the wrong way round.

Go to EDIT OP1 page 6/6 KNOB 3 and set RateScale to 3. Do the same on the other operator that's carrying sound. Play the two notes again.

The two notes should now be about the same length. If the high one still rings on, go higher.

8.4 An order to do this in

All of this is optional, all of it is fiddly, and it's easy to spend an hour here and end up with a sound that's worse. So, in order of how much they repay the effort:

1. RateScale, on whichever operators are producing sound. Notes get shorter as they get higher.
2. KeyVelocity on your modulators. Playing harder gets brighter, which over MIDI means note by note and on the FM-1's own keys means moving Global Velocity.
3. Keyboard scaling with a minus curve on the right, on your modulators. High notes get purer.
4. A ModSens, only if you're already using the LFO's AMD and want the shimmer rather than the throb.

You do not have to do any of it, and a patch with none of it set is a perfectly good patch. It just won't change as you move around the keyboard, and after this chapter you'll be able to hear that it isn't.

Chapter 9 - The arpeggiator

Hold a chord down and let the machine play it one note at a time.

An arpeggio is a chord played as a sequence rather than all at once. An arpeggiator is a machine that does it for you: you hold the notes, it decides the order and the timing, and it keeps going until you let go.

It is the easiest way to sound like you can play, and everyone who has ever owned a synthesizer has spent an evening with one. It is also genuinely useful, because it lets you hear a patch moving without having to play anything - which makes it a good tool for the sound design in chapters three to five, not just a toy.

ARP - press it, ARP lights, and the display goes to the arpeggiator settings. Hold down one or more keys and it plays them automatically. The keys light as each note is triggered. Press ARP again to stop and go back to Home.

> **One at a time**
>
> The arpeggiator and the sequencer cannot both be on. If the sequencer is running and you press ARP, the screen says “Disable SEQ to enter” and nothing happens. Turn the sequencer off first.
>
> This works the other way round too, and the message is the mirror image of it. If a button seems to have stopped doing anything, this is usually why.

9.1 Page 1 - the notes

ARP page 1/2 - Mode, Octave, Rate and Tempo, on knobs 1 to 4.

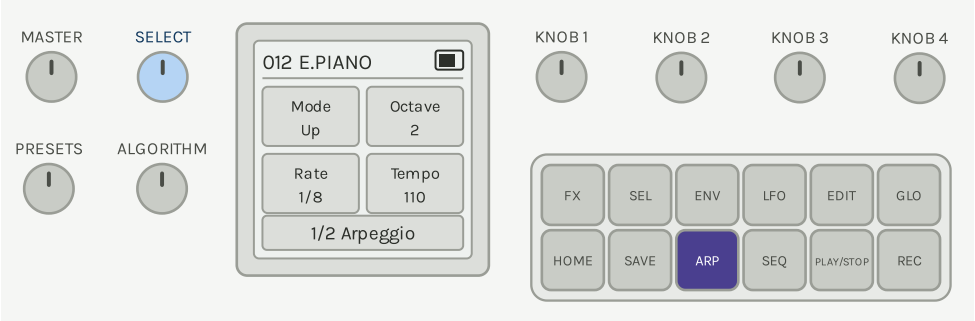


Figure 9.1 - Arpeggiator page 1.

Mode (KNOB 1, seven of them) - the order the held notes are played in. Described below.

Octave (KNOB 2, 1-4) - how far up the pattern spreads. At 1 you get the notes you're holding. At 4 the arpeggiator plays them, then the same notes an octave up, then two octaves up, then three, before starting again. It is the difference between a figure and a cascade.

Rate (KNOB 3, ten divisions) - how busy the pattern is. The choices are 1/1, 1/2, 1/4, 1/4T, 1/8, 1/8T, 1/16, 1/16T, 1/32 and 1/32T, so the difference between one note per beat, two, four, or a triplet subdivision. The triplets start at 1/4, and there is no 1/1T or 1/2T to go hunting for. A firmware update added 1/1 and 1/2 to the list.

Tempo (KNOB 4, 30-300 BPM) - how fast the music is. Set tempo once, then find the busyness with Rate. They are easy to confuse and they do different jobs. This tempo is the arpeggiator's own. The sequencer in chapter ten has a separate one, and changing either leaves the other where it was.

9.1.1 The seven modes

Up plays from the lowest note to the highest and starts again. Down does the reverse. Inclusive goes up and then back down, playing the top and bottom notes twice as it turns round. Exclusive does the same bounce but doesn't repeat the notes at the turns. Random picks notes in no order. Order plays them in the order you pressed the keys. Repeat plays each note twice before moving on.

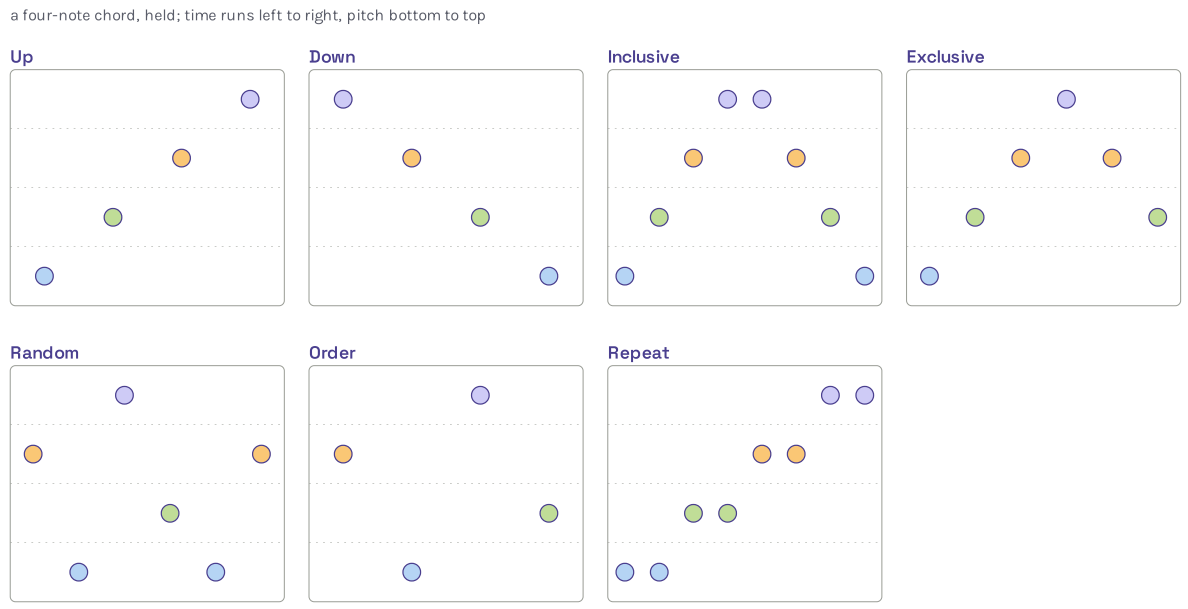


Figure 9.2 - The seven modes. Inclusive repeats the notes at the turns; Exclusive does not.

Up is the default choice for almost everything. Order is the interesting one, because it's the only mode where the sequence depends on how you played the chord rather than what the notes are - press the same four keys in a different order and you get a different pattern.

9.2 Page 2 - the feel

ARP page 2/2 - Gate, Swing, Latch and Sync, on knobs 1 to 4.

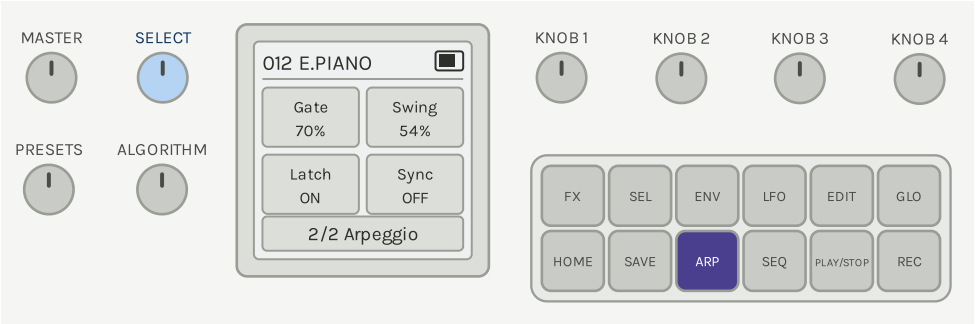


Figure 9.3 - Arpeggiator page 2.

Gate (KNOB 1, 20-100%) - how much of each slot the note actually fills. Low gate is spiky and separated; high gate runs each note into the next.

Swing (KNOB 2, 50-75%) - 50% is dead straight, every note evenly spaced. As you go up, alternate notes get delayed, so the rhythm starts to lope rather than march. A few percent above 50 is the amount that sounds like a human being. Above about 66% it sounds like a deliberate stylistic decision, which it is.

Latch (KNOB 3, On or Off) - with Latch on, the arpeggio carries on after you take your hands off the keys. Press a new set of keys and it switches to those. Turn Latch off to stop it, and it finishes the cycle it is on before stopping rather than cutting off where it stands.

Sync (KNOB 4, On or Off) - follow an external MIDI clock instead of the Tempo setting. It matters if you're playing along to something else, and does nothing at all if you aren't.

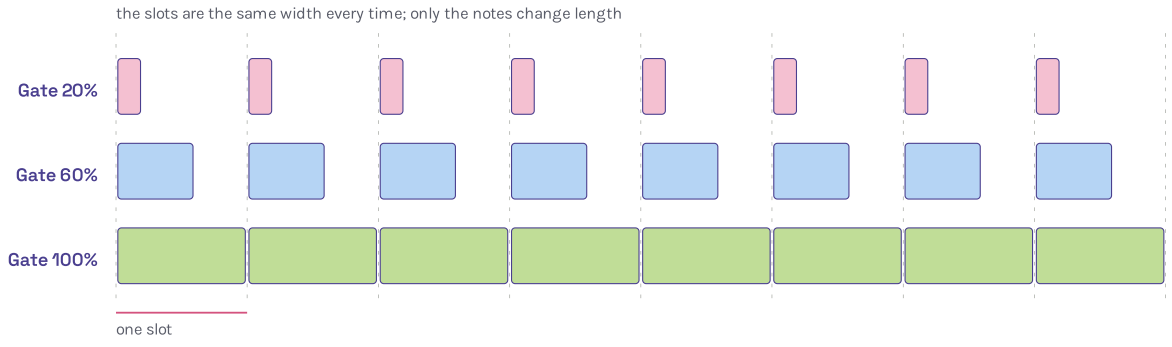


Figure 9.4 - Gate changes how long each note lasts, not how often they come.

Gate matters less than you'd expect on a sound with a long release, because the release fills the gaps anyway. Which is worth knowing when a patch that sounded crisp on its own turns to mush under the arpeggiator.

9.3 Latch, and why it's the useful one

Latch is the setting that makes the arpeggiator useful rather than entertaining. Latch a chord, take your hands away, and now both hands are free to work through the operator pages while the sound keeps playing. Every change you make to the patch is audible immediately, on a moving part, without you having to hold a key down with one hand.

Use it as a workbench

Load your electric piano from chapter four. Press ARP. Set Mode to Up, Octave to 2, Tempo to somewhere around 110, and on page 2 turn Latch on.

Hold a four-note chord, then let go. It keeps playing.

Now press EDIT and go to OP2's output level, and turn it slowly up and down while the arpeggio runs. You are hearing what that knob does, continuously, on a real musical figure, with both hands free. This is a much better way to learn a patch than playing one note at a time.

These settings belong to the instrument, not the patch

The arpeggiator is the odd one out. Everything else you have edited in this book belongs to the patch, so it changes when you change preset. These follow you from preset to preset instead.

They do not survive the power going off, though. Nothing does, except what you have saved to a preset.

Chapter 10 - The sequencer

Sixteen steps, and the keyboard stops being a keyboard again.

An arpeggiator plays the notes you're holding. A sequencer plays the notes you played earlier. That's the whole difference, and it's the difference between an accompaniment and a composition.

The FM-1's is a sixteen-step pattern sequencer, and there are sixteen patterns, so there's rather more here than the size of the instrument suggests.

> **One at a time, again**
>
> The sequencer and the arpeggiator cannot both be on. Press SEQ while ARP is running and the screen says “Disable ARP to enter”. Chapter nine said the same thing from the other side.

10.1 Switching it on

SEQ - SEQ lights, PLAY/STOP comes on orange, and the display goes to the sequencer pages.

And the keyboard stops being a keyboard. The lower row of keys becomes sixteen step indicators: lit means there's a note on that step, unlit means the step is empty, and the one currently playing blinks. Nothing on the screen tells you where you are, so watch the keys.

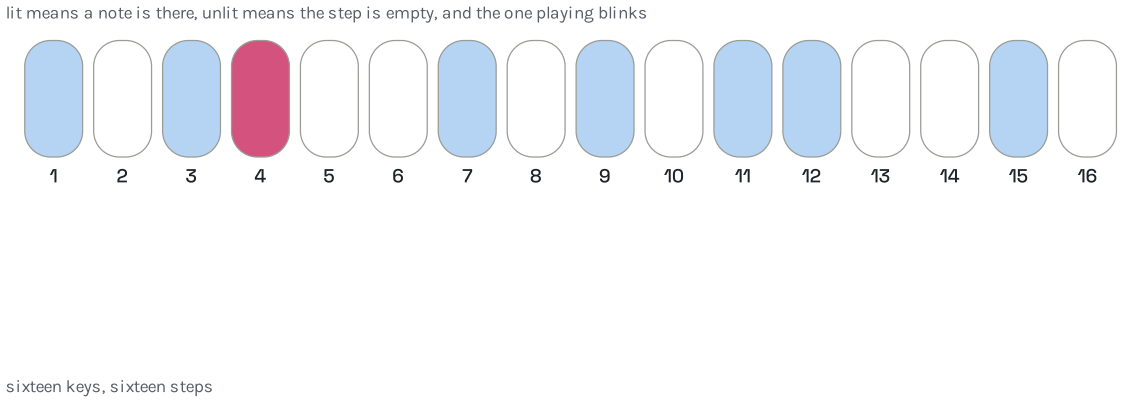


Figure 10.1 - The keys as sixteen steps.

This is the second time the keyboard has changed job - chapter two did it for EDIT - and it's the same idea: the FM-1 has more functions than it has surfaces, so the keys mean different things at different times. The lit keys are how you tell.

Press SEQ again to switch the whole thing off. All the lit buttons go out and you're back at the main screen.

10.2 Recording

1. Press REC. It lights red.
2. Play notes. Each one goes into the next step along, and that step's key lights.
3. Press REC again to stop. You're now in playback mode.

That's the whole basic operation, and it's worth noticing what it doesn't involve: no metronome, no counting in, no playing in time. You're filling sixteen slots one at a time, at your own pace, and the sequencer sorts out the timing afterwards.

Play more than sixteen notes and it wraps round to step 1 and overwrites what is there, without warning you. The lit key is the only thing that tells you it has happened, which is the other reason to keep an eye on the keys.

10.2.1 Notes that last longer than one step

On the step you're on, hold the note down and turn SELECT, then let go when you reach the step you want it to end on. A long note recorded this way doesn't wipe out notes that were already there.

That last part is the useful bit. It means you can lay down a bass line and then add a held chord over the top of it without losing the first one.

10.2.2 Rests, and changing your mind

While recording, hold REC down and turn SELECT to move around the steps. Hold keys on the step you've landed on to add or replace notes there. Leave a step empty and you get a rest.

So the sequencer has two ways of working: play it in from the beginning and accept what you get, or walk to a particular step and fix it. The second is slower and it is how most sequences actually end up sounding right.

10.3 Playing it back

PLAY/STOP - the button lights green and the sequence runs through its filled steps, with the current step blinking as it goes. Press it again to stop, and it goes back to orange.

Both PLAY/STOP and REC only do anything when the sequencer is on. Press either without it and the screen says “Enable SEQ first”.

10.4 The three pages

Three pages this time rather than two, which makes the sequencer the deepest mode on the instrument.

10.4.1 Page 1 - which pattern, and how much of it

SEQ page 1/3 - Pattern, Clear, Chain and Step, on knobs 1 to 4.

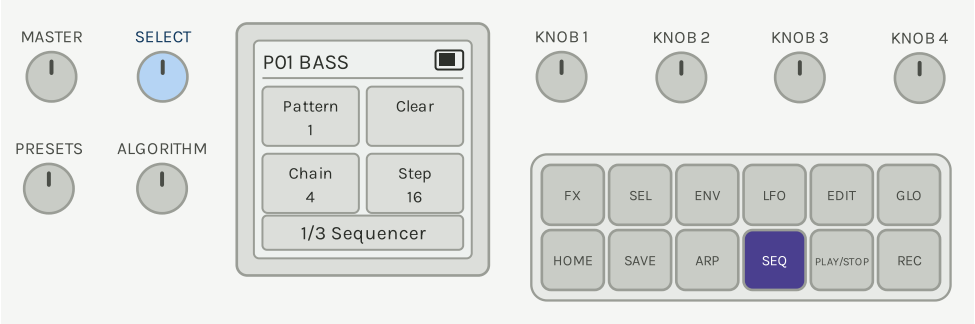


Figure 10.2 - Sequencer page 1.

> **Pattern** (KNOB 1, 1-16) - which of the sixteen patterns you're working on.

> **Clear** (KNOB 2) - empties the current pattern. It asks first, and offers two answers: KNOB 3 clears the step you are on, KNOB 4 clears the whole pattern, KNOB 2 backs out.

> **Chain** (KNOB 3, 1-16) - how many patterns to play one after another, starting from the one you're on. Section 10.6 is about this.

> **Step** (KNOB 4, 1-16) - how many steps the pattern loops round. Set it to 12 and you get a twelve-step loop, whatever is recorded in steps 13 to 16. It affects recording as well as playback, so at 12 a take wraps at 12. Shortening a pattern does not destroy the steps beyond the new end, so you can shorten it, hear it, and lengthen it again with everything intact.

10.4.2 Page 2 - what it sounds like and how fast

SEQ page 2/3 - Voice, Rate, Tempo and Gate, on knobs 1 to 4.

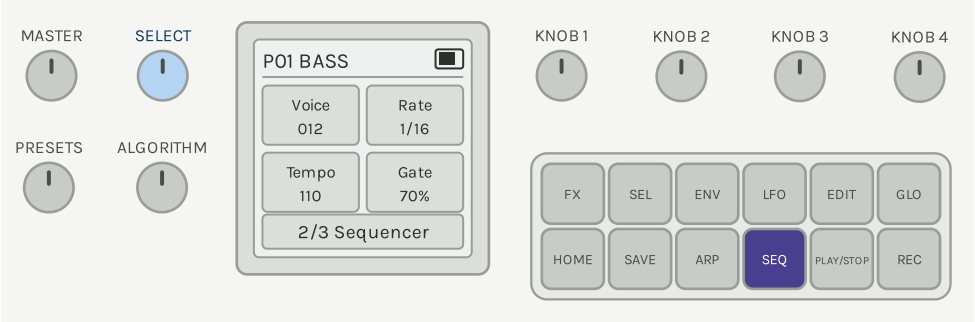


Figure 10.3 - Sequencer page 2.

> **Voice** (KNOB 1, 1-128) - which preset this pattern plays. Each of the sixteen patterns gets its own.

> **Rate** (KNOB 2) - the note division, as on the arpeggiator. You can change it while recording, and it applies to the step you're on. Rate really is stored per step, so a pattern's length is the sum of its steps rather than a fixed grid. Change one step from 1/32 to 1/1 and the whole pattern gets longer.

> **Tempo** (KNOB 3, 30-300 BPM) - how fast.

> **Gate** (KNOB 4, 20-100%) - how much of each step the note fills. Same control, same effect, as chapter nine.

A rate held per step rather than across the whole pattern is unusual at this price, and it is the best thing about this sequencer.

10.4.3 Page 3 - feel

SEQ page 3/3 - Swing, Sync and Transpose, on knobs 1 to 3.

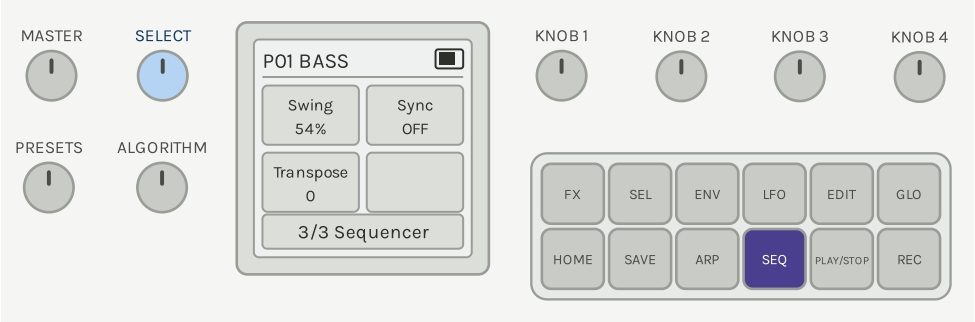


Figure 10.4 - Sequencer page 3.

Swing (KNOB 1, 50-75%) - how much the rhythm lopes. As on the arpeggiator: a few percent is a human being, a lot is a decision.

Sync (KNOB 2, On or Off) - follow an external MIDI clock instead of the Tempo setting.

Transpose (KNOB 3, -24 to +24) - shifts the whole sequence in semitones, two octaves either way.

> **Three transposes, three scopes**
>
> There are now three ways to transpose the instrument and they do not reach the same things. This one moves the pattern only. The octave buttons used with ALGORITHM move what you play live only. The one on the patch Globe page, EDIT then GLO, moves both, keeping whatever interval you have set between them.

10.5 Sixteen patterns, each with its own sound

There are sixteen patterns and they're independent. Each one can be given its own voice, on page 2, and all sixteen patterns and their voices are written to flash automatically - there is no save step.

> **The PRESETS knob stops working**
> With the sequencer on, the PRESETS knob does nothing. The sound comes from the pattern's own Voice setting instead.
>
> Which is sensible once you know it and baffling if you don't, because PRESETS is the knob you've been using since chapter one and it is the last thing you'd expect to go dead.

10.6 Chaining patterns together

Sixteen steps is a bar or two. A song is longer than that, and Chain is how you get there.

Set Chain on page 1 to a number and the sequencer plays that many patterns in succession, starting from whichever one you're on, then goes back to the beginning.

So four patterns of sixteen steps, chained, gives you sixty-four steps of music - and because each pattern carries its own voice, the sound can change as it goes. That is a great deal more than a sixteen-step sequencer sounds like it ought to manage.

Chain is a length rather than a list, and it wraps. Set Chain to 4 and start on pattern 14 and you get 14, 15, 16, 1. Start the same setting on 15 and you get 15, 16, 1, 2. So where you start is a performance decision, not a setting.

> **Build something in four patterns**
> Record a bass line into pattern 1. Move to pattern 2 with KNOB 1 and record a variation of it. Do the same for 3 and 4, changing one thing each time.
>
> Now go back to pattern 1, set Chain to 4, and press PLAY/STOP.
>
> Give patterns 3 and 4 a different voice on page 2 and play it again. The arrangement changes without you touching anything while it runs.

Chapter 11 - The FM-1 and other things

It isn’t a toy that happens to have a socket, it’s a member of a family that has been going since 1983.

Everything up to here has treated the FM-1 as a self-contained object: 27 keys, some knobs, a speaker. This chapter is about what happens when you plug it into something, and it ends with the single most useful fact in the book.

Three connections, all at once

There are three MIDI routes into the FM-1, and all three work at the same time. You don't choose between them.

USB, over the USB-C socket, which is the same socket that charges it. Plug it into a computer and it appears as a standard USB MIDI device with no drivers and no setup.

Bluetooth, as BLE MIDI, to a phone, tablet, Mac or PC that supports it. This is the one people don't expect on an instrument this cheap, and it means a tablet running a keyboard app can play it with no cable at all. It is on by default. HOME lights while Bluetooth is on, flashing while it is searching and solid once something has connected, and a long press on HOME switches it off.

And a 3.5 mm MIDI IN socket on the back, which takes a 3.5 mm-to-DIN adapter and connects to ordinary MIDI hardware.

> **No MIDI out socket**
>
> The socket on the back is MIDI in only. There is no MIDI out socket, so anything the FM-1 transmits leaves over USB or Bluetooth instead. Nor is there much of it: the FM-1 sends the notes you play and nothing else - no knob movements, no program changes, no clock - so it will not serve as a controller for anything beyond the notes themselves.
>
> The audio output expects a standard TRS headphone plug. The manual warns that other kinds of connector may produce noise and may fail to mute the internal speakers - so if you plug into an amplifier and the little speakers keep playing as well, that's the cable, not a fault.

What it listens to

Notes, with velocity. A note-on with a velocity of zero is treated as a note-off, which is the standard behaviour and matters only if you're writing software.

That first point settles half of chapter eight's open question. Whatever the FM-1's own silicone keys do or don't sense, velocity arriving over MIDI is real and the synthesis engine responds to it. If you want the velocity settings on operator page 4 to do their work, playing the FM-1 from a full-size keyboard is the way to be certain of it.

Pitch bend, over whatever range you've set. That range is on the GLO button's page 1, as `Pitch Up` and `Pitch Dn`, each 1 to 24 semitones, set separately. Two semitones each way is the conventional setting. Twelve each way is an octave, which is showing off but occasionally right.

Control change messages, which adjust parameters - principally the effects. There is a separate `Effect Chn` setting on the GLO button's page 1, 1 to 16, deciding which MIDI channel those control messages are listened to on, independently of the channel the notes come in on.

Program change, 0 to 127, to switch voices remotely. Note that this gives you 128 program numbers for 128 presets, offset by one: program change 0 selects preset 001, and program change 127 selects preset 128. If your sequencer numbers program changes from 1, everything will be one preset out.

> ### Two channels, and where the table is
>
> The MIDI channel the keyboard listens on is `Midi Chn` on the GLO button's page 1, and can be ALL or 1 to 16. `Effect Chn` sits beside it and is set separately.
>
> The manual refers to an official MIDI response table for the full list of which control change number moves which parameter, and does not print it. M-VAVE do publish it, on the download page rather than the product page, as a document called MIDI CONTROL. It is worth having: effects are on their own channel, channel 2 by default, with four control changes each in chain order, and it is accurate. One trap it flags itself - control change 1 on the effect channel moves the filter type rather than the modulation wheel, so a controller with its wheel on channel 2 will quietly reshape your filter.

What it sends

Over USB or Bluetooth, since there is no socket for it: notes, with velocity, from the keyboard, the arpeggiator or the sequencer. That is the whole list.

Which is a small thing that turns out to be useful. Latch an arpeggio, or start a sequence, and the FM-1 will play those notes into a computer or another synthesizer as well as playing them itself. It is a perfectly good little note source, and if you write in a DAW, the sequencer on this thing is a pleasant way to come up with a part.

It does not send its knob movements, so you cannot record yourself twiddling. In fact it sends nothing at all except the notes you play - no control changes, no program changes when you turn PRESETS, no clock.

Forty years of free sounds

Here is the fact this chapter exists for.

The FM-1 accepts voice banks over MIDI SysEx, and the format it accepts is the standard Yamaha DX7 bank: 32 voices at a time. When a bank arrives, a screen appears asking which slot to put it in - A, B, C or D - and you choose with the knobs.

The four slots are quarters of the one 128-preset list, so a bank sent to C lands in presets 65 to 96 and comes back on the PRESETS knob like anything else. The voices arrive with their names intact.

Several video reviews report that voices imported into B, C or D cannot be reached afterwards. That is a misunderstanding rather than a fault: a reviewer expecting a DX7's arrangement looks at preset 1, hears the factory sound, and concludes the bank has vanished, when it is sitting thirty-two slots further along.

The DX7 came out in 1983, sold in enormous numbers, and was programmed by more or less everybody. People have been making banks of sounds for it, and giving them away, for over forty years. There are archives of thousands of them online, free, in a file format that has not changed. They were made for a synthesizer that cost as much as a car.

All of them load into this.

That is why the algorithms in chapter five are the DX7's thirty-two algorithms, in the DX7's order, and why the operator envelopes have four rates and four levels instead of the four simple stages every other synthesizer uses. It isn't imitating a DX7. It's the same instrument, made small and cheap and battery-powered.

Before you fill all four banks

Importing replaces voices that are already there, and the factory sounds are what's already there. A bank fills one quarter of the preset list - A is 1 to 32, B is 33 to 64, C is 65 to 96, D is 97 to 128 - so an import into B costs you thirty-two factory sounds and leaves the other ninety-six alone.

The reset from chapter two is not the way back. It restores a bank to whatever was last imported into it, so once you have imported into B, resetting B gives you the imported bank again rather than M-VAVE's originals. Nor can you take a copy first: the FM-1 receives voices over MIDI but will not send them.

Recovery is possible, and no longer means going near M-VAVE's Chinese-language updater. The factory set has been recovered as four standard DX7 bank dumps, one per bank, at `github.com/KingParamount/fm1-factory-presets`. Send them back to the instrument with Dexed, or with the browser-based editor at `fm1-editor.com`.

So fill a bank, by all means. Just decide which bank you are willing to lose before you start, rather than after.

Also expect a fair proportion of any forty-year-old bank to be unusable. They were made for a full-size, five-octave, velocity-sensitive keyboard, often for a specific record, and a lot of them are one thin sound that made sense in one arrangement. The hit rate is low and the ones that hit are extraordinary.

What to do with all this

The first eight chapters were about building sounds from nothing, which is the slow way and the way you actually learn. This chapter hands you a shortcut that skips all of it.

Use both. Load a bank, find a sound in it that does something you can't explain, and then open it up in the operator pages and work out what makes it do that. Every patch in every one of those archives is a worked example, made by somebody who knew what they were doing, and you now know how to read them.

Appendix A - Control changes

How to work out which CC moves what, without a table that goes stale.

The FM-1 takes control changes for every effect parameter, which means a sequencer can automate the effects even though the instrument itself will never record you turning a knob.

M-VAVE publish the full list. It is on their download page rather than the product page, in a document called MIDI CONTROL, and it is worth downloading because it is accurate and this appendix deliberately is not exhaustive.

> **Why the numbers are not printed here**
>
> A control change number that has quietly changed between firmware versions does not announce itself. Nothing goes wrong on screen, no error appears, and no sound stops. A knob you did not mean to touch simply moves, and you are left unable to tell whether the book is wrong or you are.
>
> That is a worse failure than a menu screen that has been rearranged, where at least you can see the mismatch. So what follows is the shape of the scheme, which is unlikely to change, rather than a copy of a table that might.

A.1 The arithmetic

Effect parameters arrive on their own MIDI channel, separate from the notes. It defaults to channel 2 and is set on the GLO button's page 1, on KNOB 2.

On that channel the effects take four control changes each, in chain order, starting from zero. Filter has 0 to 3, Reverb 4 to 7, Delay 8 to 11, Distortion 12 to 15, Chorus 16 to 19, Phaser 20 to 23.

Within each group of four, the first is always the on/off switch: send it zero to switch the effect off and anything above zero to switch it on. The remaining three are that effect's own parameters, in the order they appear on screen when you select it.

So the whole thing reduces to one line: the *n*th effect in the chain starts at CC 4*n*, counting from zero, and the first of its four is the switch. Reverb is second, so it starts at CC 4. Chorus is fifth, so it starts at CC 16.

Values run 0 to 100, matching what the screen shows, with two exceptions. Filter Cutoff is displayed as a frequency rather than a number. Filter Q moves in ten steps rather than a hundred, so only multiples of ten do anything.

A.2 Three things to know before you send anything

The effect channel steals CC 1. On that channel the modulation wheel message moves the filter type instead of the modulation. If your controller has a mod wheel and you have left it on channel 2, it will quietly reshape the filter every time you touch it. Move the notes to another channel, or move the effects.

Program changes are offset by one. Program change 0 selects preset 001 and program change 127 selects preset 128. If your sequencer numbers program changes from 1, everything will be one preset out.

Nothing comes back. The FM-1 accepts all of this and transmits none of it: no control changes, no program changes, no clock, and no way to read a patch out of it. Whatever you set from a sequencer, the sequencer has to remember, because the instrument will not tell you what it is doing.